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VOLUME OF TARGET TISSUE OR KIT
INCLUDING SAME****Publication Classification**(51) **Int. Cl.**

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(57)

ABSTRACT

The present invention provides a pharmaceutical composition for treating obesity, the composition including: one or more viruses selected from the group consisting of yellow fever virus, herpes zoster virus, and rubella virus; or a genetic material coding for a protein derived from these viruses. Preferably, the pharmaceutical composition is a vaccine composition. The composition provides a reduction in target tissues, preferably tissues containing adipocytes, or an effect that leads to the death of adipocytes.

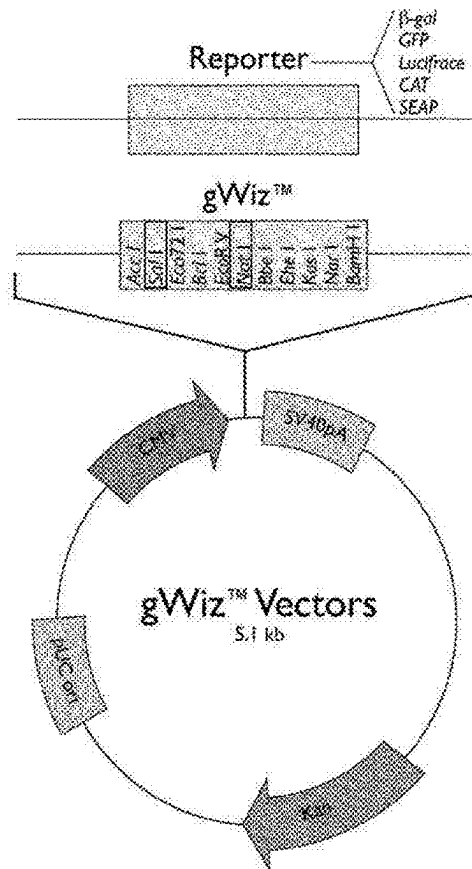
Specification includes a Sequence Listing.

FIG. 1

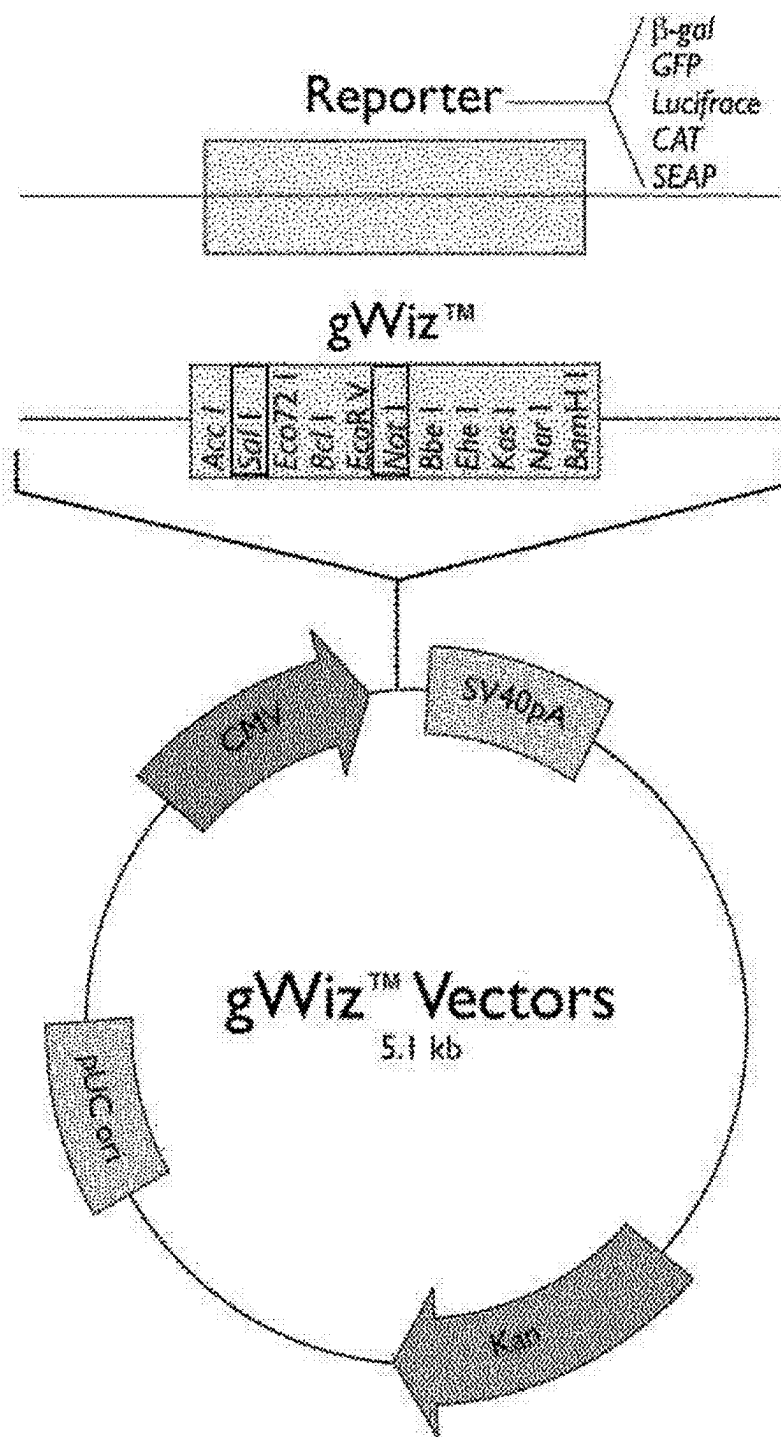


FIG. 2

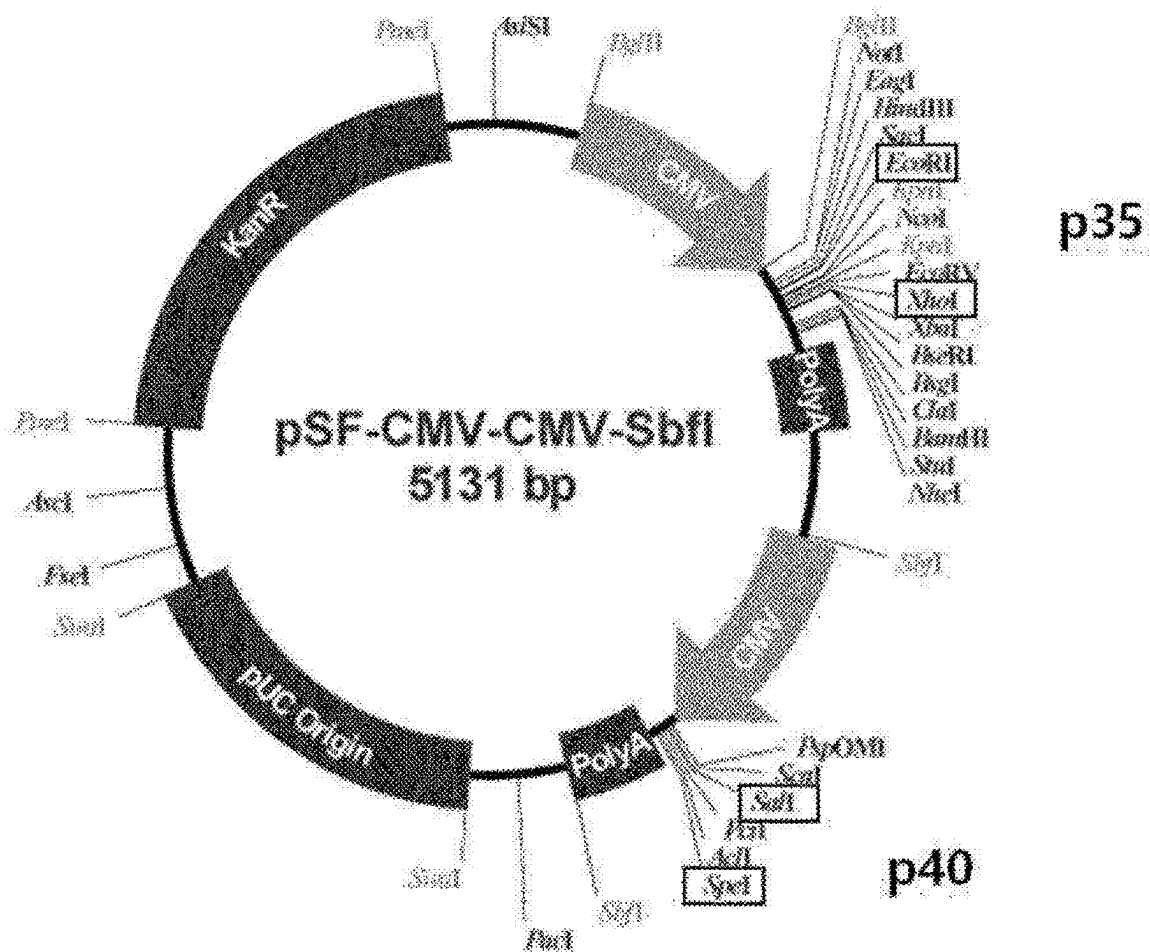


FIG. 3

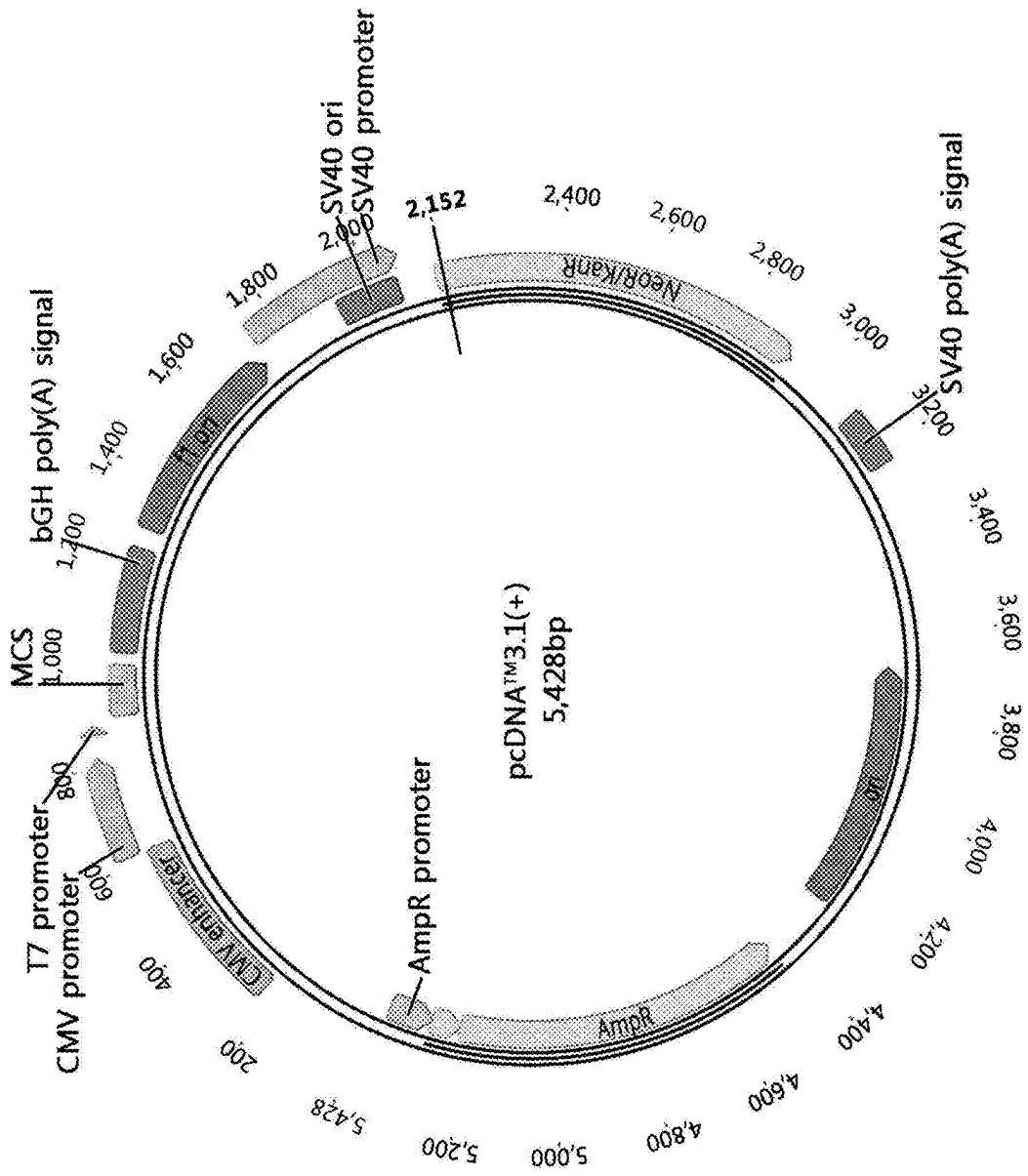


FIG. 4

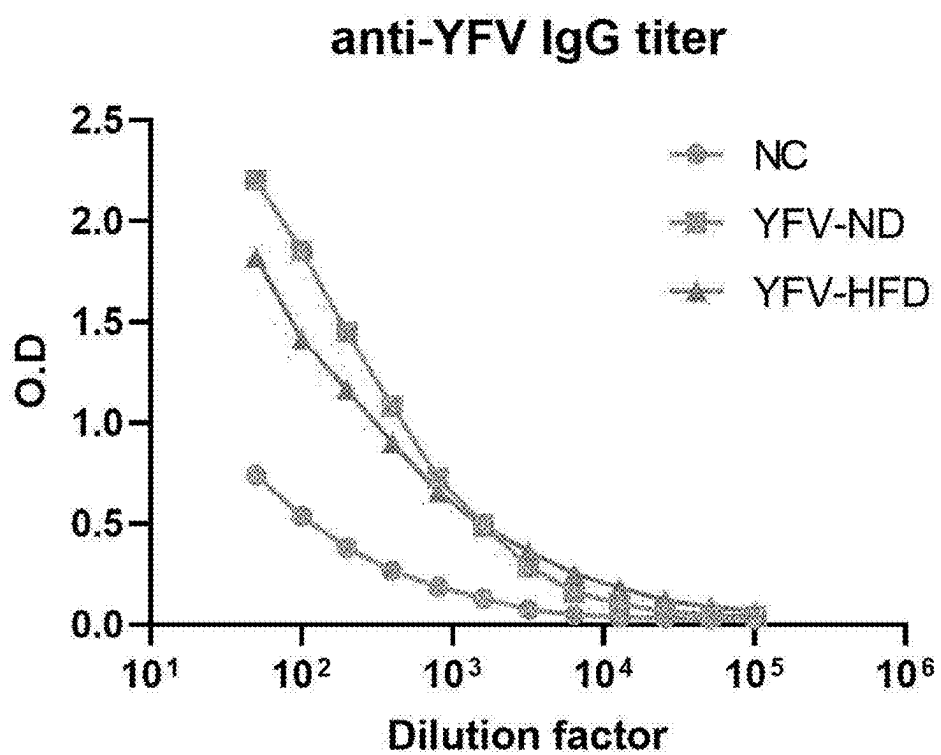


FIG. 5

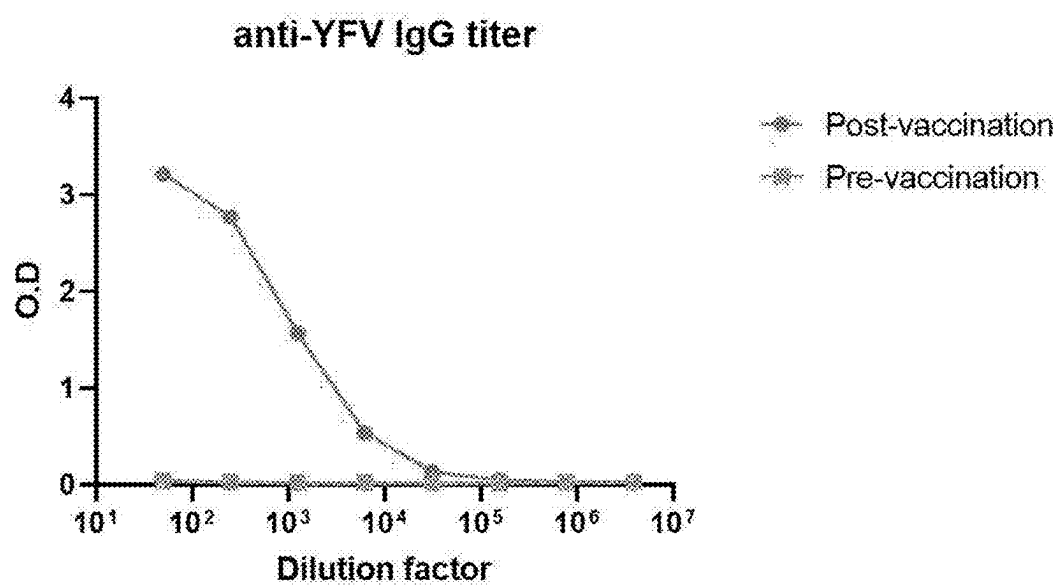


FIG. 6

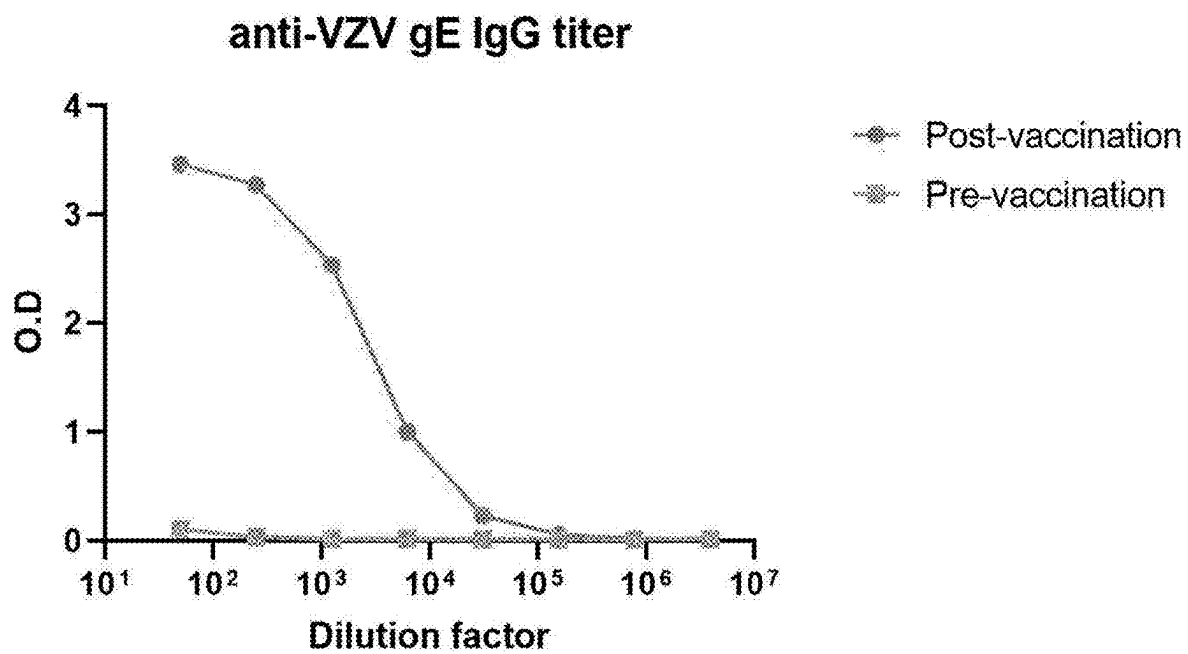


FIG. 7

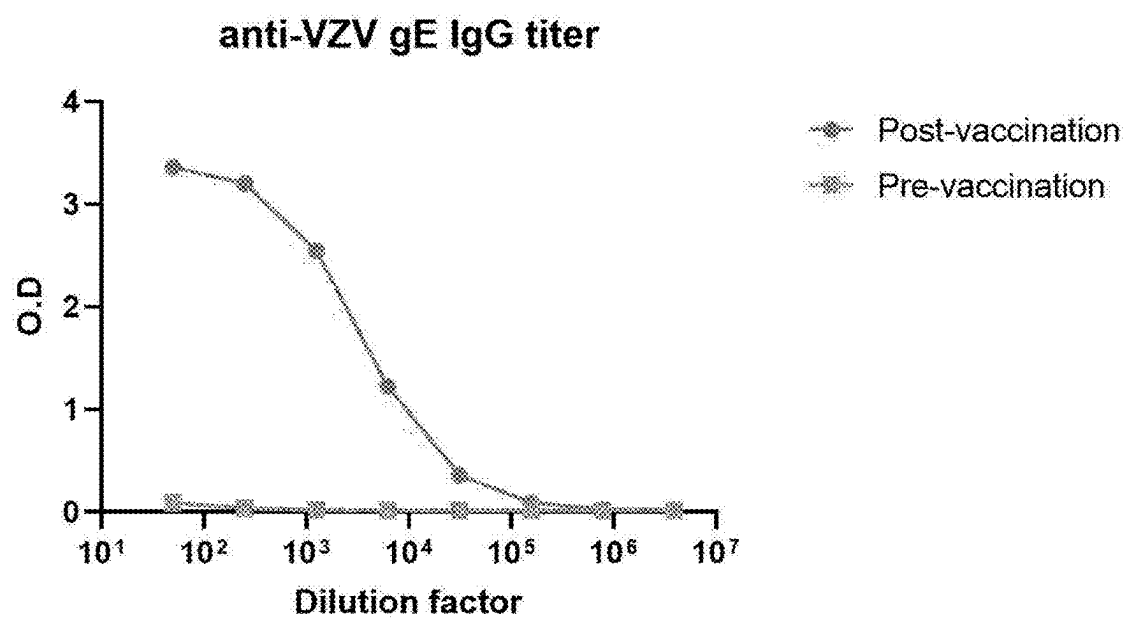


FIG. 8

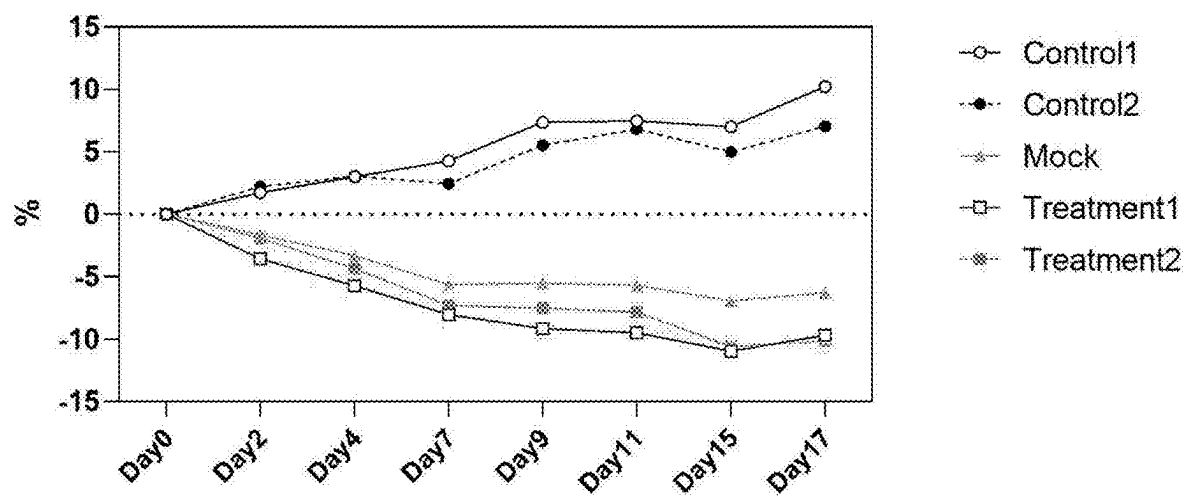


FIG. 9

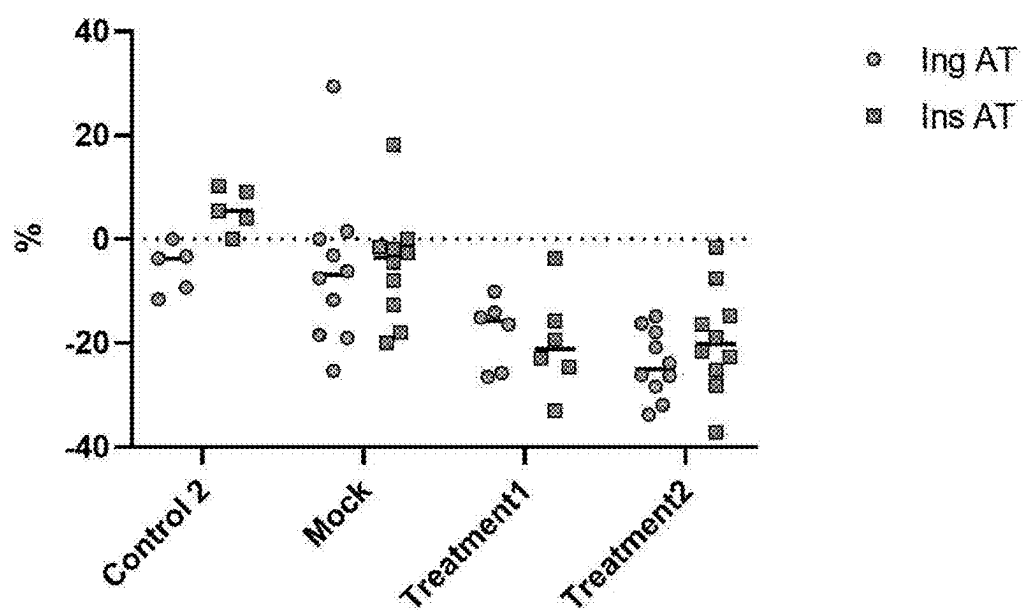


FIG. 10

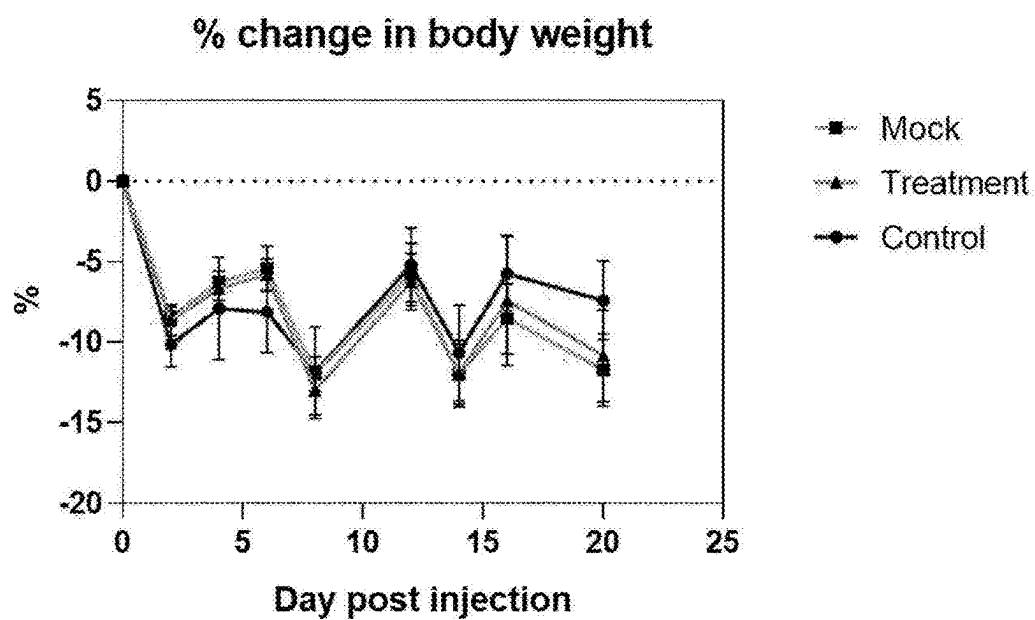


FIG. 11

% change in adipose tissue weight

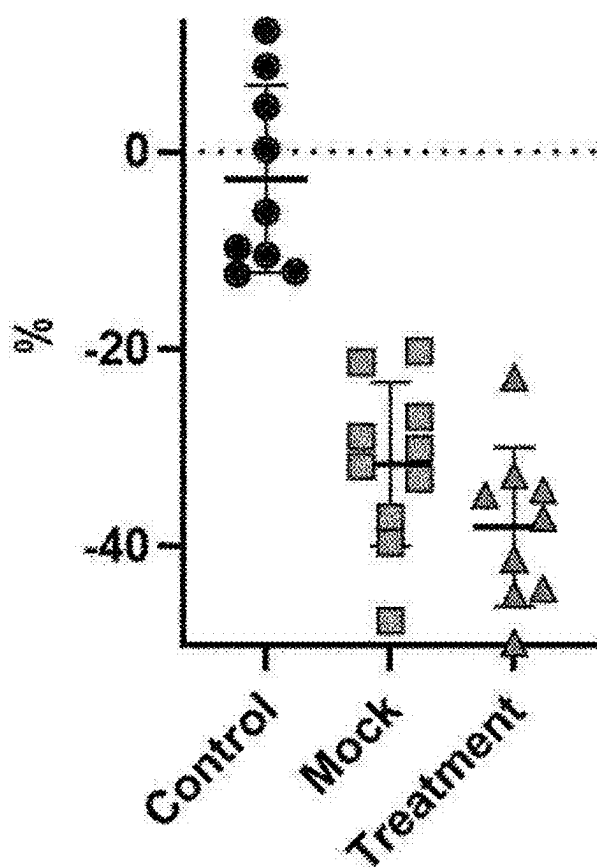


FIG. 12

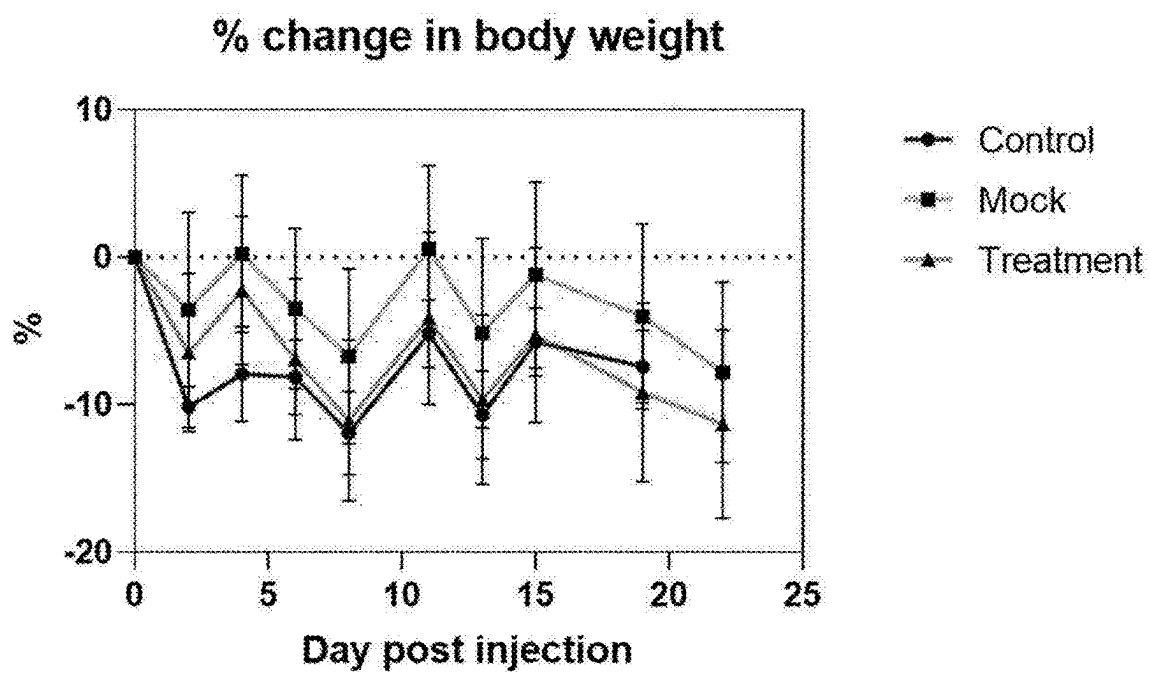


FIG. 13

% change in adipose tissue weight

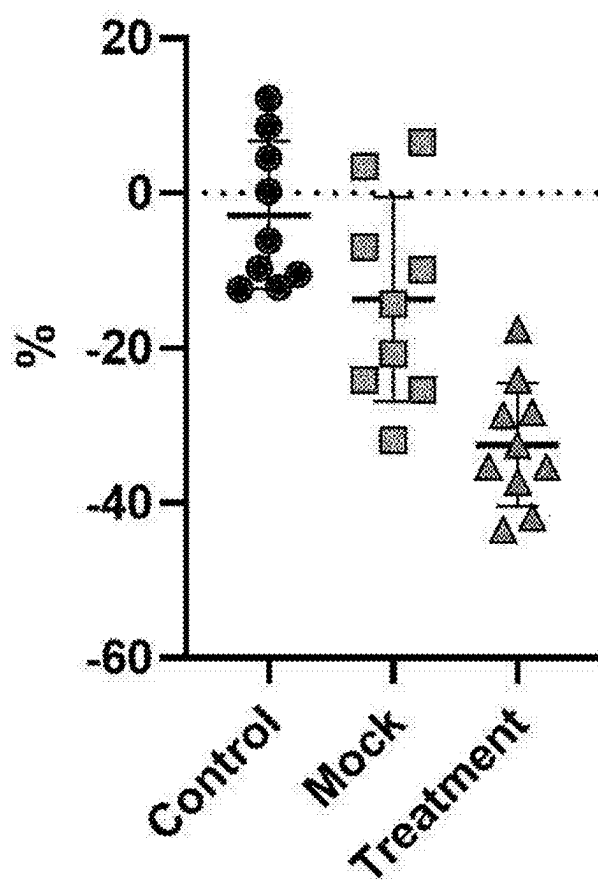


FIG. 14

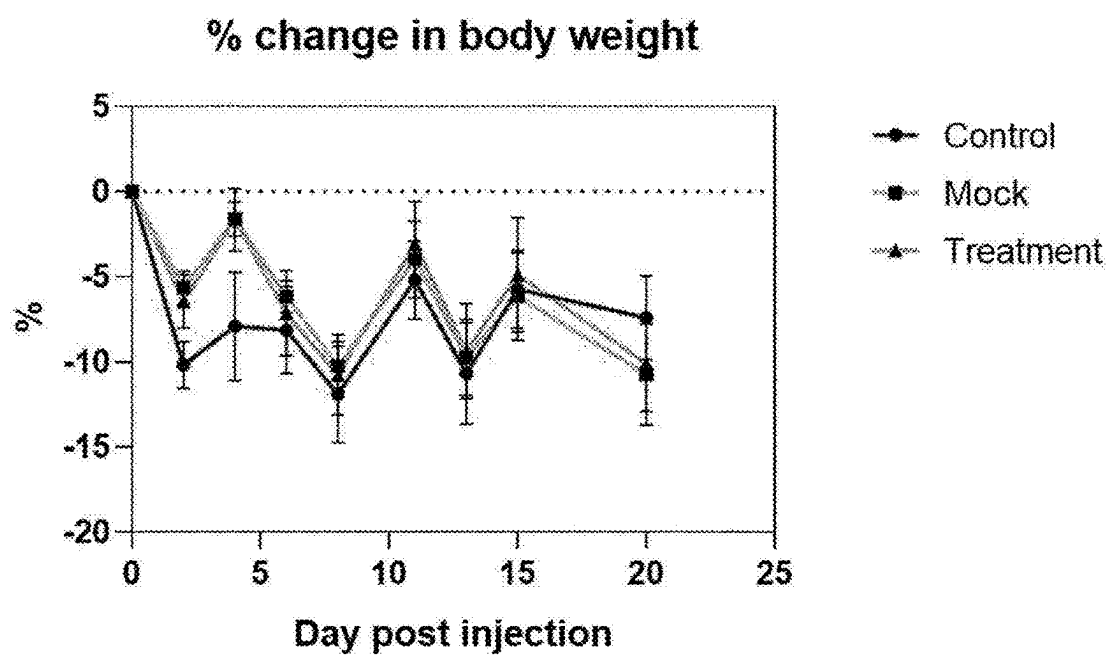
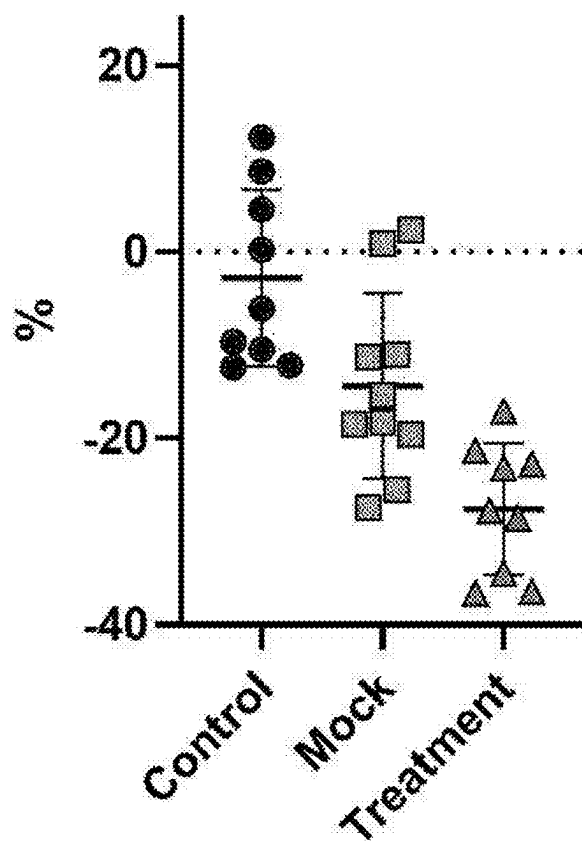


FIG. 15

% change in adipose tissue weight



COMPOSITION FOR REDUCING SIZE OR VOLUME OF TARGET TISSUE OR KIT INCLUDING SAME

TECHNICAL FIELD

[0001] The present application claims the priority based on Korean Patent Application No. 10-2022-0050367 filed on Apr. 22, 2022, and the entire contents disclosed in the description and drawings of the corresponding application are incorporated in the present application. The present invention relates to a composition for reducing the size or volume of a target tissue, a kit containing the same, a use thereof, or a method for treating or improving obesity using the same. More particularly, it pertains to a composition that reduces the size or volume of a target tissue by killing mast cells, inhibiting mast cell accumulation, or reducing mast cells. Additionally, it includes a kit containing the same, a use thereof, or a method for treating or improving obesity using the same.

BACKGROUND ART

[0002] Obesity refers to a 'condition in which an excessive amount of body fat is accumulated in the body' rather than simply being overweight, and is recognized as a risk factor that increases the incidence of various chronic diseases such as metabolic diseases including diabetes, cardiovascular diseases, cancer and the like. Obesity, defined as excessive accumulation of fat, is accompanied by loss of metabolic, endocrine and immune functions of adipose tissue, and pathological remodeling of this adipose tissue is attracting attention as a pathophysiological factor of major metabolic diseases.

[0003] As a drug treatment for treatment of obesity, 'Orlistat', a lipolytic enzyme inhibitor, is used, which plays a role of discharging some of fat in the body out of the body, and side effects include diarrhea and steatorrhea, and the like. In addition, in case of drugs for treatment of obesity, not only do they have many side effects, but also they can decompose fat in an undesired site of a woman's body. At present, there is no appetite suppressant that can be safely used, and in case of severely obese patients with complications, bariatric surgery for the gastrointestinal tract may be helpful.

[0004] Meanwhile, while technologies to prevent the formation of fat have been developed, the field of reducing the size of already formed adipocytes and adipose tissue still requires significant research.

DISCLOSURE

Technical Problem

[0005] In order to solve the above problem, the present invention is to provide a new method or pharmaceutical composition which can reduce a size or volume of target tissue by administering into target tissue.

[0006] A problem to be solved by the present invention is to provide a new composition for reducing target cells that helps the immune system attack target cells expressing antigen proteins, leading to the apoptosis of the target cells while not affecting other cells in undesired areas.

[0007] Additionally, the invention seeks to provide a novel use of the composition that can reduce, decrease, or induce apoptosis in the desired cells or tissues by administering foreign antigens.

Technical Solution

[0008] One embodiment of the present invention provides a composition for the treatment of obesity using a virus or its fragments, an obesity treatment vaccine, a therapeutic method, or a use for treating obesity. One embodiment of the present invention provides a novel pharmaceutical composition (preferably a vaccine composition) for reducing tissue size or volume and a pharmaceutical composition (preferably a vaccine composition) for improving, suppressing or treating obesity, which administers a virus or a genetic material encoding the virus to a target tissue, and when an antigen gene is expressed in a cell constituting the target tissue by the administered genetic material, a pre-existing immunity acts on it to kill the cell. More specifically, the present invention aims to provide a pharmaceutical composition for the prevention or treatment of obesity containing a virus or genetic material coding for the virus, a composition for removing subcutaneous adipocytes, a pharmaceutical composition for reducing the size or volume of tissue (preferably a vaccine composition), and pharmaceutical compositions for improving, inhibiting, or treating obesity. Preferably, the composition may include a vaccine formulation.

[0009] The virus of the present invention includes the virus itself, viral-derived proteins, and genetic material coding for viral-derived antigens, and may include fragments, variants, or derivatives thereof. Preferably, the genetic material can be understood to include genetic material coding for viral-derived antigens. In particular, the virus may include one or more selected from the group consisting of yellow fever virus, varicella-zoster virus, and rubella virus. Furthermore, viruses belonging to higher taxa or higher orders that include these viruses may also be included in the present invention, as long as they do not hinder the objectives of the invention.

[0010] The inventors of the present invention has confirmed that tissue composed by the cells can be reduced by killing cells constituting a target site using an immune response, thereby completing the present invention. In particular, it has been confirmed through an experiment that the present invention can achieve effective removal and/or reduction of adipocytes, and through this, body shape correction and obesity treatment effects can be shown.

[0011] One embodiment of the present invention provides a novel use of a pharmaceutical composition comprising a genetic material, which administers a virus or a composition comprising a virus to a target tissue, and causes an antigen gene to be expressed in cells constituting the target tissue by the administered genetic material, thereby reducing the size of fat, particularly subcutaneous adipocytes. The use provides a use in which, after the composition is administered, a pre-existing immune response acts to kill the corresponding cells, thereby reducing the size or volume of the tissue, and a use in improving, suppressing, or treating obesity.

[0012] The inventors of the present invention confirmed that the cells constituting the target area can be killed by using an immune response by a virus, thereby reducing the tissues, and thus completed the present invention. In particular, it was confirmed through experiments that the pres-

ent invention can effectively remove and/or reduce target cells, thereby exhibiting body shape correction and obesity treatment effects. In particular, when the virus was a yellow fever, shingles, and/or rubella virus, it was advantageous in achieving the effects of the present invention.

[0013] One embodiment of the present invention is a pharmaceutical composition for administering to a target tissue to reduce the tissue, wherein the composition may include genetic material encoding a virus, preferably at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses. One embodiment of the present invention provides a use of the genetic material encoding the virus for target cell killing, target tissue reduction, and more specifically, provides a use of the genetic material for local adipose tissue reduction. Preferably, the genetic material encoding the virus comprises at least one virus selected from the group consisting of yellow fever virus, shingles virus, and rubella virus; or an epitope thereof. One embodiment may provide a use of the virus or an epitope thereof for target cell killing or target tissue reduction. The tissue may be understood as a tissue including adipocytes. The immune action of the present invention may reduce the size of adipocytes or adipose tissue. For example, the adipocytes or adipose tissue may be more preferably subcutaneous adipose tissue. The composition can induce a decrease, reduction or death of adipocytes, particularly adipocytes distributed in subcutaneous adipocytes.

[0014] The inventors of the present invention provide a composition or kit that has fewer side effects than existing fat removal surgeries, drugs, etc., and can expect a body shape correction effect by targeting only a local target area (for example, adipose tissue of a specific area). The present invention provides a pharmaceutical composition that acts locally on a target area and reduces the size or volume of a target tissue using immunotherapy. The present invention can preferably provide a pharmaceutical composition, vaccine composition, or therapeutic vaccine composition that induces the death of adipocytes, particularly subcutaneous adipocytes. The present invention can provide a pharmaceutical composition, vaccine composition, or therapeutic vaccine composition that acts locally on a target area and improves, suppresses, or treats obesity using immunotherapy.

[0015] The terms “Target tissue or target site” used in the present description means a cell group having a structure and a function for reducing a size or volume. The “figure correction or body shape correction” means having an externally slim and balanced effect by killing or reducing cells constituting tissue (for example, adipose tissue) accumulated in an undesired site to reduce a size or volume of tissue. “Locally act” used in the present description means bring about an effect of cell death or reducing a size or volume of tissue within a radius of 10 cm, 9 cm, 8 cm, 7 cm, 6 cm, 5 cm, 4 cm, 3 cm, 2 cm, 1 cm, based on an injection administration site, and preferably, it may bring about the effect within a range of 3 cm.

[0016] When a subject is exposed to a virus-derived antigen, immunocytes functionally perform functions of phagocytosis and antigen presenting. In one embodiment, the pharmaceutical composition of the present invention may be administered after an immune environment is created by intentionally or unintentionally infiltrating a foreign substance such as a protein antigen or an epitope thereof, or the like to induce an immune response. Through another

example, the pharmaceutical composition may be repeatedly administered in the body. In other example, the genetic material comprised in the pharmaceutical composition of the present invention is administered to induce pre-existing immunity, and then with repeated administration of the composition, adipocytes presenting the antigen by a substance secreted by an immune action such as a monocyte, neutrophil, natural killer cell, one kind of lymphocytes, cytotoxic T cell immunocyte, physiologically active substance, antibody, complement, or the like in the body die, and therefore, the size or volume of the adipose tissue may be reduced. The composition may be used as a use of figure correction or body shape correction, and it may be used as a use of body shape correction on a local site. In addition, the composition may be administered to muscle tissue and used as an effect of skin wrinkle improvement such as Botox, a use of figure correction through muscle contraction, and the like. Moreover, it may be used for removing benign or malignant tumors, or removing warts on skin.

[0017] The pharmaceutical composition comprises a genetic material encoding one or more a virus-derived antigen, and the genetic material includes not only DNA and/or RNA, but also a viral vector. Also, it comprises antisense nucleotide, mRNA, ssRNA (Single-Stranded RNA), cDNA, and the like, associated with the genetic material. The antigen(s) may also comprise an epitope of an antigen that elicits an immune response.

[0018] The inventors of the present invention have confirmed that when a virus-derived antigen, preferably a virus-derived antigen whose association with obesity is unknown, more preferably the genetic material of the virus-derived antigen, and even more preferably the DNA or RNA of the virus-derived antigen, is administered, an excellent effect of reducing the number of adipocytes, reducing the size of adipose tissue, or inhibiting fat accumulation is exhibited. In particular, it has been confirmed that among viruses, yellow fever, shingles, and/or rubella viruses, which were previously unknown to have any association with obesity, have an excellent effect of reducing adipocyte size.

[0019] The virus-derived genetic material may be expressed as a viral peptide or a complete protein or a portion thereof. For example, non-limitative examples of the virus include Retroviridae (for example, human immunodeficiency virus, for example, HIV-1; Picornaviridae (for example, poliovirus, hepatitis A virus; enterovirus, human coxsackie virus, rhinovirus, echovirus); Calciviridae (for example, strain causing gastroenteritis); Togaviridae (for example, equine encephalitis virus, rubella virus); Flaviridae (for example, dengue virus, encephalitis virus, yellow fever virus); Coronaviridae (for example, corona virus); Rhabdoviridae (for example, vesicular stomatitis virus, rabies virus); Filoviridae (for example, Ebola virus); Paramyxoviridae (for example, parainfluenza virus, mumps virus, measles virus, respiratory syncytial virus); Orthomyxoviridae (for example, influenza virus); Bunyaviridae (for example, Hantaan virus, Bunga virus, and phlebovirus and Nairo virus); Arenaviridae (hemorrhagic fever virus); Reoviridae (for example, reovirus, orbivirus and rotavirus); Birnaviridae; Hepadnaviridae (hepatitis B virus); Parvoviridae (parvovirus); Papovaviridae (papillomavirus, polyomavirus); Herpesviridae (herpes simplex virus (HSV) 1 and 2, varicella-zoster virus, cytomegalovirus (CMV), herpes virus; Poxviridae (variola virus, vaccinia virus, poxvirus); and Iridoviridae (for example, African swine fever virus).

Preferably, it may comprise Rubivirus genus, more preferably Rubella virus (also called rubella virus); preferably Flavivirus genus, more preferably Yellow fever virus; preferably Varicellovirus genus, more preferably varicella-zoster virus. When the virus was administered, the effect of reducing tissue size was excellent. In addition, when the above virus was administered, there were few side effects and it was safe. The virus was easy to formulate. Even if a small amount of the virus is administered, it can stimulate pre-existing immunity, thereby quickly bringing about a fat decomposition effect.

[0020] Preferably, the genetic material encoding the virus may comprise a nucleic acid molecule having a nucleic acid sequence consisting of a polynucleotide selected from the group consisting of SEQ ID NOs: 1 to 6. Alternatively, the genetic material may comprise a nucleic acid molecule encoding any one or more peptides selected from the group consisting of SEQ ID NOs: 12 to 14.

[0021] One embodiment of the present invention can provide a pharmaceutical composition, a vaccine composition, or a therapeutic vaccine composition comprising a nucleic acid molecule comprising a polynucleotide selected from the group consisting of SEQ ID NO: 2, SEQ ID NO: 4, and SEQ ID NO: 6. Wherein SEQ ID NO: 2, SEQ ID NO: 4, and SEQ ID NO: 6 can include a polynucleotide sequence encoding at least one virus (or a protein derived from a virus) selected from the group consisting of yellow fever virus, varicella-zoster virus, and rubella virus following a capping sequence, a 5'-UTR, a Kozak sequence. A 3'-UTR sequence, a poly A Tail, and a primer can be linked following the polynucleotide sequence encoding the virus (or a protein derived from a virus), unless otherwise specified, and these are connected in the 5' to 3' direction. See FIGS. 16 to 18.

[0022] In addition, the genetic material may include a nucleic acid sequence consisting of a polynucleotide selected from the group consisting of SEQ ID NOs: 1 to 6; or a polynucleotide having a nucleic acid sequence having at least 85% sequence homology with the nucleic acid sequence and capable of reducing target tissue size or volume, decomposing fat, or treating, improving, or preventing obesity (i.e., the present invention can achieve the purpose). The sequence homology may include a sequence homology of 85% or more, 90% or more, 95% or more, or 99% or more. Or, the genetic material may include a polynucleotide having a nucleic acid sequence having at least 85% sequence homology with a nucleic acid molecule encoding any one or more peptides selected from the group consisting of SEQ ID NOs: 12 to 14 and capable of reducing target tissue size or volume (i.e., the present invention can achieve the purpose). The sequence homology may include a sequence homology of 85% or more, 90% or more, 95% or more, 99% or more, or 100%.

[0023] The composition may further comprise a genetic material encoding at least one cytokine selected from the group consisting of IL-12, IL-2, IL-4, IL-5, IFN- γ , IL-10, IL-1, IL-6, INF-alpha, INF-beta, TNF-alpha, and TNF-beta, and it is preferable to comprise IL-12 to further maximize the immunostimulatory effect by an antigen expressed by the genetic material. For example, the IL-12 may comprise a DNA sequence consisting of SEQ ID NO: 7 and/or SEQ ID NO: 8.

[0024] In a preferred embodiment, the viral genetic material can be included in a plasmid (e.g., gWizTM vector) that

enables the expression of a gene corresponding to a structural protein of the virus in an animal cell.

[0025] In a preferred embodiment, the virus can include a yellow fever virus, for example, but not limited to, the yellow fever virus can include 17D strain. The plasmid can include a nucleic acid molecule encoding the virus, or a fragment, variant, or analog thereof. For example, it can include SEQ ID NO: 1 or SEQ ID NO: 2; or a fragment, variant, or analog thereof.

[0026] The nucleic acid molecule encoding the virus may comprise a nucleic acid molecule encoding a structural protein. For example, it may comprise a nucleic acid molecule encoding a protein of SEQ ID NO: 12, or a fragment, variant or analog thereof.

[0027] In a preferred embodiment, the viral genetic material can be included in a plasmid (e.g., gWizTM vector) that enables protein expression in an animal cell of a gene corresponding to a structural protein of a Rubella virus. The Rubella virus can be used without limitation, and for example, the Rubella virus can include, but is not limited to, RA27/3 strain. The gene encoding the virus can include a gene encoding spike glycoproteins. The plasmid can include a nucleic acid molecule encoding the virus or a fragment, variant, or analog thereof. For example, it can include SEQ ID NO: 5 or SEQ ID NO: 6; or a fragment, variant, or analog thereof.

[0028] Examples of the spike glycoproteins may include E2-E1 heterozygotes. For example, the nucleic acid molecule encoding the protein of SEQ ID NO: 14 or a fragment, variant or analog thereof may be included.

[0029] In one preferable embodiment, the composition may comprise a gene corresponding to a structural protein of a rubella virus gene in a plasmid enabling protein expression in an animal cell (for example, gWizTM vector). The rubella virus may be used without limitation, and for example, the rubella virus may include RA27/3 strain, but not limited thereto. The gene encoding the virus may comprise a gene encoding spike glycoproteins. The example of the spike glycoproteins may include E2-E1 heterozygotes. For example, it may comprise a nucleic acid molecule encoding protein of SEQ ID NO: 5 or a fragment, variant or analogue thereof.

[0030] In one preferable embodiment, the composition may comprise a gene corresponding to a structural protein of a varicella-zoster virus gene in a plasmid enabling protein expression in an animal cell (for example, gWizTM vector). The varicella-zoster virus may be used without limitation, and for example, the varicella-zoster virus may include Oka strain, but not limited thereto. The above plasmid may comprise a nucleic acid molecule encoding a virus or a fragment, variant, or analog thereof. For example, it may comprise SEQ ID NO: 3 or SEQ ID NO: 4; or a fragment, variant, or analog thereof.

[0031] The nucleic acid molecule encoding the virus may comprise a nucleic acid virus encoding gE protein, a surface protein. For example, it may comprise a nucleic acid molecule encoding protein of SEQ ID NO: 13 or a fragment, variant or analogue thereof.

[0032] Here, the term 'fragment, variant or analog thereof' may be understood understood as meaning a part of a proteins, peptides, nucleic acids, nucleic acid molecules or genes which achieves the purpose intended by the proteins, peptides, nucleic acids, nucleic acid molecules or genes of the present invention, but may have a sequence homology of

85% or more, 90% or more, 95% or more or 99% or more. For example, the meaning of comprising the fragment, variant or analogue may be understood as meaning that if it includes a nucleic acid molecule composed of any one or more polynucleotides selected from the group consisting of SEQ ID NOs: 1 to 6, it is not excluded from the scope of the present invention.

[0033] The virus may have an excellent effect of death of target tissue, preferably, adipocytes, and reducing a size of tissue. In addition, the effect may be quickly exhibited, and there are few side effects. Moreover, there may be little risk of deterioration even if stored for a long time as an injection.

[0034] In other embodiment, the composition may further enhance an effect of adipocyte death, or reducing target tissue, by comprising an immune enhancer capable of enhancing a cellular immune response. The composition may further comprise a genetic material encoding an immune enhancer capable of improving an immune enhancing effect, for example, an auxiliary genetic material encoding interleukin 12 (IL-12). The auxiliary genetic material may be provided by inserting a coding region for example, into pSF-CMV-CMV-Sbfl vector.

[0035] The composition of the present invention may be used with not only the cytokine, a gene encoding the same or other immunoadjuvant, but also a pharmaceutical carrier or excipient as widely used in the art. The composition may be formulated for human or veterinary use, and administered in various routes. As an administration route, an oral, transdermal, intramuscular, peritoneal, intravenous, subcutaneous, or intranasal route may be used, but not limited thereto. More preferably, a transdermal, intramuscular, peritoneal or subcutaneous route may be used. In addition, the composition may be administered by any device or route in which an active material can move to a target cell, and the administration method is not particularly limited, as long as the composition of the present invention or the genetic material comprised in the composition can be delivered to target tissue or a cell constituting the tissue. For example, it may be formulated and provided as an injection, microneedle patch, or the like. For example, when used as an injection, the injection may be prepared using an aqueous solvent such as physiological saline solution, Ringer solution and the like, a non-aqueous solvent such as plant oil, higher fatty acid ester (e.g., oleic acid ethyl, etc.), alcohols (e.g., ethanol, benzyl alcohol, propylene glycol, glycerin, etc.) and the like, within a range that does not impede the effect of the genetic material, and may comprise a pharmaceutical carrier such as a stabilizer for preventing deterioration (e.g., ascorbic acid, sodium hydrogen sulfite, sodium pyrosulfite, BHA, tocopherol, EDTA, etc.), an emulsifier, a buffer for adjusting pH, a preservative for preventing microbial growth (e.g., phenyl mercury nitrate, thimerosal, benzalkonium chloride, phenol, cresol, benzyl alcohol, etc.), and the like. When used as a microneedle patch, the microneedle may include both insoluble and soluble microneedles.

[0036] In addition, the genetic material and/or auxiliary genetic material comprised in the composition may be delivered into the body according to a common gene delivery method in the art. As a non-limitative example, gene delivery methods such as physical delivery methods such as electroporation or gene gun and chemical delivery methods such as lipid-gene complex (Lipid-DNA complex: Lipoplex), polymer-gene complex (Polymer-DNA complex:

Polyplex), liposome, dendrimers, nanoparticles, or other appropriate transfer vectors may all be used.

[0037] The composition of the present invention is administered in a pharmaceutically effective dose. The term “pharmaceutically effective dose” refers to an amount sufficient for a composition to exhibit an effect of treatment, improvement or prevention, or to exhibit a vaccine effect, and in addition, it means an amount sufficient to not cause side effects or a severe or excessive immune response. The composition of the present invention may be administered with a genetic material due to characteristics of the purpose of inducing an immune response and inducing cell death, and it is preferable to administer it 4-8 times, preferably, 5 times at a 2-3 day interval. When administered as described above, an effect of decrease or reduction of tissue may be obtained by cell death of the target site. The composition may have a difference in the cell death effect as it is administered in large amounts or multiple times, and may be used by adjusting an appropriate dosage for each site.

[0038] Other embodiment of the present invention provides a method for reducing or decreasing a size of volume of tissue, comprising administering a genetic material encoding a genetic material encoding an antigen, preferably a virus, such as yellow fever, shingles or rubella virus, a vector comprising the genetic material, or a composition comprising thereof into target tissue of a subject. The present method may be effective in killing cells that make up subcutaneous adipocyte tissue, reducing fat weight, or reducing cell size. The i) method; or ii) the genetic material encoding a virus or virus-derived protein, a vector comprising the genetic material, or a composition comprising thereof, or a kit comprising thereof may be used as a use for improving or treating obesity of a subject, a use for wrinkle improvement, a use for muscle reduction, a use for removing or reducing benign or malignant tumors, a use for removing warts, and the like, and may be used as a use for removing fat pockets under eyes for a cosmetic purpose, and may be used as a use for alleviating dark circles through this.

[0039] The method may further comprise composing an immune environment in the body prior to administering the genetic material, a vector comprising the genetic material, or a composition comprising thereof into target tissue. The composing an immune environment in the body may comprise administering a foreign substance capable of inducing an immune response. The foreign substance may include any substance as long as it uses protein as an antigen without limitation.

[0040] One embodiment of the present invention may be provided as a use for treatment, improvement or prevention of various indications, and may be provided as a cosmetic purpose for improvement of appearance, as well.

[0041] One embodiment of the present invention can be provided as a vaccine for treating obesity. The therapeutic vaccine can be defined as a vaccine in that it utilizes an immune response in the body, and can be defined as a therapeutic vaccine in that it affects fat, adipocytes, or adipose tissues that have already been created to reduce their size or kill cells. Therefore, the vaccine for treating obesity can be understood as a concept that is distinct from a preventive vaccine that simply suppresses the formation of adipocytes or suppresses the accumulation of fat.

[0042] Other example of the present invention provides a kit, comprising (a) a vaccine composition for reducing the size or volume of a target tissue, comprising at least one

virus selected from the group consisting of yellow fever, shingles, and rubella viruses; a protein derived from said virus; or genetic material encoding said virus or protein, preferably comprising at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses or genetic material encoding said virus or protein, and (b) an administration guide for administration of the vaccine. The kit may further include (c) a composition comprising a protein antigen or an epitope thereof for immune environment composition prior to administering the (a) composition. Herein, the (c) may be interpreted as at least one pre-vaccine or composition for inducing an immune response.

[0043] The ‘pre-vaccine’ mentioned herein can be understood as a vaccine composition administered before the ‘pharmaceutical composition for improving, suppressing, or treating obesity’ of the present invention, or the ‘composition for reducing target tissue size or volume’ is administered to obtain pre-existing immunity. The pre-vaccine may include an antigenic substance identical to the antigenic substance included in the ‘pharmaceutical composition for improving, suppressing, or treating obesity’ of the present invention, or the ‘composition for reducing target tissue size or volume’, but is not limited thereto. For example, (a) the ‘pharmaceutical composition for improving, suppressing, or treating obesity’, or the ‘composition for reducing target tissue size or volume’, comprising genetic material encoding a virus, preferably a yellow fever, shingles, and/or rubella virus, may include at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses, and (c) may also include at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses. As another example, (a) a ‘pharmaceutical composition for improving, suppressing, or treating obesity’, or a ‘composition for reducing target tissue size or volume’, comprising genetic material encoding a virus, preferably at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses, may comprise at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses, and (c) may comprise a vaccine that is administered in infants to induce the retention of antibodies in the body.

[0044] The composition (a) above may further include a genetic material encoding a cytokine, and a material that can be used to increase the immune effect of the composition (a) above may be included without limitation. The genetic material encoding the cytokine may be replaced with a protein expressed by the genetic material. The cytokine may include at least one selected from the group consisting of IL-12, IL-2, IL-4, IL-5, IFN- γ , IL-10, IL-1, IL-6, INF-alpha, INF-beta, TNF-alpha, and TNF-beta, and in addition to the cytokines above, cytokines known in the art may be included without limitation.

[0045] The kit of the present invention, may comprise at least one additional agent as defined in the present description in relation to a pharmaceutical composition, an antimicrobial agent, a DNase inhibitor, a RNase inhibitor, a solubilizing agent, and the like. The kit may for example, consist of at least one kit parts (for example, a container). For example, each container may be a syringe, a prefilled syringe, vial, bottle, jar, sealed sleeve, envelope or pouch, tube or blister package or any other suitable form in which the container is configured to prevent premature mixing of elements. Each of different elements may be provided indi-

vidually, or some of the different elements may be provided together (i.e., in the same container). The kit may also comprise an administration guide having information on administration, administration method and dosage of any component.

[0046] Another embodiment of the present invention provides a syringe filled with a composition for improving, suppressing, or treating obesity, preferably a vaccine composition, comprising genetic material encoding a virus or a viral protein. Another embodiment of the present invention provides a syringe filled with a pharmaceutical composition for reducing target tissue size or volume, preferably a vaccine composition, comprising genetic material encoding a virus or a viral protein. Preferably, a syringe filled with a composition for improving, suppressing, or treating obesity, preferably a vaccine composition, comprising at least one virus selected from the group consisting of yellow fever, varicella-zoster virus, and rubella viruses; or a genetic material encoding the virus-derived protein; or a vaccine composition for reducing target tissue size or volume. The syringe may further contain a genetic material encoding a cytokine. The cytokine may include at least one selected from the group consisting of IL-12, IL-2, IL-4, IL-5, IFN- γ , IL-10, IL-1, IL-6, INF-alpha, INF-beta, TNF-alpha, and TNF-beta.

[0047] One embodiment of the present invention is to provide a pharmaceutical composition for improving, suppressing, or treating obesity, comprising genetic material encoding at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses; or a protein derived from said virus.

[0048] The pharmaceutical composition can reduce the size or volume of a target tissue.

[0049] The genetic material may be a polynucleotide encoding at least one protein selected from the group consisting of a structural protein of yellow fever virus, glycoprotein E of shingles virus, and E2 and E1 proteins of rubella virus.

[0050] The polynucleotide may be DNA or RNA, and preferably may be DNA or mRNA.

[0051] The genetic material may comprise a nucleic acid molecule composed of one or more polynucleotides selected from the group consisting of SEQ ID NOS: 1 to 6.

[0052] The pharmaceutical composition may further comprise a genetic material encoding a cytokine.

[0053] The cytokine may be one or more selected from the group consisting of IL-12, IL-2, IL-4, IL-5, IFN- γ , IL-10, IL-1, IL-6, INF-alpha, INF-beta, TNF-alpha, and TNF-beta. For example, the genetic material may comprise a polynucleotide DNA sequence represented by SEQ ID NOS: 7 and 8, respectively.

[0054] The genetic material may induce an intracellular immune response.

[0055] The composition acts locally, and the composition may be applied after at least one antigen for creating an immune environment is first applied to the subject.

[0056] The antigen may include a virus, protein, or an epitope thereof introduced from outside the human body. Preferably, the genetic material may be administered to the target tissue by oral administration, transdermal, intramuscular, peritoneal, intravenous, subcutaneous, or nasal injection, or by electroporation, gene gun, liposome, dendrimers, nanoparticles, or transfer vector.

[0057] The dosage of the genetic material included in the composition may be 0.1 to 1000 ug/site per single inoculation.

[0058] The composition is administered to an individual for the purpose of reducing or shrinking the size of adipose tissue, or killing it, and the composition may be administered at least once to a site for the purpose of reducing or shrinking the size of adipose tissue, or killing it after confirming the production of antibodies against at least one of the viruses after administration of a vaccine for preventing any one or more viruses selected from the group consisting of yellow fever, shingles, and rubella viruses.

[0059] The composition may be provided as a vaccine, such as a therapeutic agent, a therapeutic vaccine, or a prophylactic vaccine.

[0060] Another embodiment may provide a kit for providing the composition. The kit may include, in addition to the composition, an administration guideline for administering the composition.

[0061] The kit may further include an additional composition comprising an antigen for creating an immune environment in the body before administering the composition. The kit may further include a genetic material encoding a cytokine.

[0062] Another embodiment may provide a pharmaceutical composition for reducing the size or volume of a target tissue, comprising at least one virus selected from the group consisting of yellow fever, varicella-zoster virus, and rubella viruses; or a genetic material encoding a protein derived from the virus.

[0063] The pharmaceutical composition may reduce the number of adipocytes or the size of adipocyte tissue, or induce death of the cell or tissue.

[0064] According to one embodiment, the present invention provides a method for reducing the size or volume of a target tissue by administering the composition to a subject in need thereof. Another embodiment provides a method of suppressing, improving, or treating obesity by administering the composition to a subject in need thereof. The subject may include, without limitation, a mammal, such as a dog, a cow, a pig, a horse, a human, and the like, and preferably a human.

[0065] In one embodiment, the method may provide a method for improving, suppressing, or treating obesity, or a method for reducing, shrinking, or killing adipose tissue, in a subject for the purpose of reducing, shrinking, or killing adipose tissue, comprising the steps of: i) administering a vaccine for preventing at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses, ii) confirming the production of antibodies by the vaccine, and iii) administering a pharmaceutical composition comprising at least one virus selected from the group consisting of yellow fever, shingles, and rubella viruses; a protein derived from the virus; or a genetic material encoding the virus or protein. The above method can repeat step iii) 1 or more times, 2 or more times, 3 or more times, 4 or more times, 5 or more times, 6 or more times, 7 or more times, 8 or more times, 9 or more times, or 10 or more times, preferably 2 or more times, 3 or more times, 4 or more times, 5 or more times, 6 or more times, 7 or more times, 8 or more times, 9 or more times, or 10 or more times, more preferably 3 or more times, 4 or more times, 5 or more times, 6 or more times, 7 or more times, 8 or more times, 9 or more times, or

10 or more times. Preferably, step iii) can be repeated at intervals of 1 to 10 days, at intervals of 2 to 8 days, or at intervals of 3 to 7 days.

[0066] The composition used in step iii) above may further comprise one or more cytokines, preferably the cytokines may comprise one or more cytokines selected from the group consisting of IL-12, IL-2, IL-4, IL-5, IFN- γ , IL-10, IL-1, IL-6, INF-alpha, INF-beta, TNF-alpha, and TNF-beta. As another alternative, the step iii) above may comprise a step of iv) administering a composition comprising one or more cytokines, preferably the cytokines may comprise one or more cytokines selected from the group consisting of IL-12, IL-2, IL-4, IL-5, IFN- γ , IL-10, IL-1, IL-6, INF-alpha, INF-beta, TNF-alpha, and TNF-beta. That is, the cytokine may be administered together with or sequentially with a composition containing the genetic material of the virus or virus-derived protein of the present invention.

[0067] One embodiment of the present invention provides a method of improving or therapeutically improving a portion of a virus selected from the group consisting of yellow fever, shingles, and rubella viruses; the viral protein; or a substance encoding the virus. Another example provides for the use of a composition comprising at least one virus selected from the group consisting of yellow fever, varicella-zoster virus, and rubella viruses; a protein derived from said virus; or genetic material encoding said virus or protein, for reducing the size or volume of a target tissue.

[0068] The above use may be used for the purpose of reducing the size or volume of a target tissue. Preferably, the target tissue may comprise a tissue comprising adipocyte cells.

[0069] The genetic material may be a polynucleotide encoding at least one protein selected from the group consisting of a structural protein of a yellow fever virus, a glycoprotein E of a varicella-zoster virus, and E2 and E1 proteins of a rubella virus.

[0070] The poly polygon may comprise DNA or RNA, and the RNA may alternatively comprise mRNA.

[0071] The genetic material may comprise a nucleic acid molecule consisting of one or more polynucleotides selected from the group consisting of SEQ ID NOS: 1 to 6.

[0072] The genetic material may be used to induce an immune response within a cell.

[0073] The composition may be administered to act locally, and the composition may be applied after at least one antigen for creating an immune environment has been first applied to the subject.

[0074] In one embodiment, the composition is administered to a subject for reducing the size, shrinking the size, or killing adipose tissue, and the composition may be administered at least once to an area for the purpose of reducing the size, shrinking the size, or killing adipose tissue after administration of a vaccine for preventing any one or more viruses selected from the group consisting of yellow fever, shingles, and rubella viruses and confirmation of antibody production against any one or more of the viruses.

Advantageous Effects

[0075] The present invention provides the effects of reducing the size or volume of a tissue in a specific targeted area. The present invention offers a new type of composition or therapeutic vaccine that can improve or treat obesity. This composition has the ability to destroy cells in a specific target tissue, leading to a reduction in the size or volume of

that tissue. By administering the composition, a body shaping effect or a body improvement effect can be obtained.

[0076] The present invention can provide a new type of obesity treatment which can solve local obesity and complete a smooth body line.

[0077] The composition for improving, suppressing, or treating obesity in the present invention is effective for selective local site reduction, allowing for the selective reduction of tissue only in the desired area, without affecting tissue in undesired areas or the cells that constitute them.

[0078] In addition, the composition of the present invention can be used for a cosmetic purpose for appearance improvement.

BRIEF DESCRIPTION OF THE DRAWINGS

[0079] FIG. 1 shows the vector map of the gWiz™ vector for inserting the genetic material of the present invention. The gene was inserted at the restriction enzyme sites Sall and NotI.

[0080] FIG. 2 shows the vector map of the pSF-CMV-CMV-Sbfl vector, which was used to insert the nucleic acid sequences corresponding to the coding regions of IL-12 p35 (GenBank: M86672.1) and p40 (GenBank: M86671.1) (p35; 127-774 bp, p40; 35-1042 bp). IL-12 p35 was inserted at the restriction enzyme sites EcoRI and XhoI in MCS site I, while IL-12 p40 was inserted at the restriction enzyme sites Sall and SpeI in MCS site II.

[0081] FIG. 3 shows the vector for in vitro transcription (IVT).

[0082] FIG. 4 presents the results of inducing pre-existing immunity by administering a YFV vaccine strain (live attenuated virus) to mice. After administering the vaccine strain three times at two-week intervals, the levels of YFV-specific IgG antibodies in the blood were measured two weeks after the last administration.

[0083] NC (negative control) indicates normal mice that received only the medium, YFV-ND indicates YFV-immunized normal mice fed a normal diet, and YFV-HFD indicates YFV-immunized obese mice fed a high-fat diet.

[0084] FIG. 5 also shows the results of inducing pre-existing immunity by administering a YFV vaccine strain (live attenuated virus) to mice. After administering the vaccine strain twice at two-day intervals, YFV-specific IgG antibody levels in the blood were measured three weeks later. Pre-vaccination refers to some individuals before YFV administration, while post-vaccination refers to YFV-immunized obese mice fed a high-fat diet.

[0085] FIG. 6 and FIG. 7 show the results of inducing pre-existing immunity by administering a VZV vaccine strain (live attenuated virus) to mice. After administering the vaccine strain twice at four-week intervals, VZV gE protein-specific IgG antibody levels in the blood were measured three weeks later. Pre-vaccination refers to some individuals before VZV administration, while post-vaccination refers to VZV-immunized obese mice fed a high-fat diet.

[0086] FIG. 8 and FIG. 9 show the results after administering a DNA vaccine for YFV. FIG. 8 measures the change in body weight before and after administration, based on the body weight just before administering the YFV genetic material (DNA). FIG. 9 illustrates the weight reduction effects on inguinal adipose tissue (Ing AT) and interscapular adipose tissue (Ins AT) due to the treatment containing YFV genetic material (DNA).

[0087] FIG. 10 and FIG. 11 show the results after administering a DNA vaccine for VZV. FIG. 10 measures the change in body weight before and after administration, based on the body weight just before administering the VZV genetic material (DNA). FIG. 11 shows the weight reduction effects on inguinal adipose tissue due to the administration of the vaccine containing VZV genetic material (DNA).

[0088] FIG. 12 and FIG. 13 show the results after administering an mRNA vaccine for YFV. FIG. 12 measures the change in body weight before and after administration, based on the body weight just before administering the YFV genetic material (mRNA). FIG. 13 illustrates the weight reduction effects on inguinal adipose tissue due to the treatment containing YFV genetic material (mRNA).

[0089] FIG. 14 and FIG. 15 show the results after administering an mRNA vaccine for VZV. FIG. 14 measures the change in body weight before and after administration, based on the body weight just before administering the VZV genetic material (mRNA). FIG. 15 shows the weight reduction effects on inguinal adipose tissue due to the administration of the vaccine containing VZV genetic material (mRNA).

[0090] FIGS. 16, 17, and 18 show the mRNA sequences of SEQ ID Nos. 2, 4, and 6, respectively. Here, 1 refers to the capping sequence, 2 refers to the 5'-UTR sequence, 3 refers to the Kozak sequence, 5 refers to the 3'-UTR sequence, 6 refers to the poly A sequence, and 7 refers to the primer sequence. The unmarked section (4) represents the GOI (Gene Of Interest) sequence. FIG. 16 includes the gene encoding the structural protein of the yellow fever virus, FIG. 17 includes the gene for the gE protein of the varicella-zoster virus, and FIG. 18 includes the gene encoding the E2-E1 protein of RuV.

MODE FOR INVENTION

[0091] Hereinafter, in order to help understanding of the present invention, it will be described in detail by examples and the like. However, the examples according to the present invention may be modified in many different forms, and the scope of the present invention should not be construed as being limited to the following examples. The examples of the present invention are provided to explain the present invention more completely to those skilled in the art. All sequences are described from the 5' to the 3' direction unless otherwise specified.

1. Preparation of Vaccine (Treatment) for Degrading Adipocytes

[0092] For the experiment, genetic material to be administered to individuals with pre-existing immunity and an immunoadjuvant that can be administered together with the genetic material were prepared as follows:

(1) Preparation of Foreign Antigens

[0093] 1) Preparation of genetic material for the yellow fever virus (Yellow fever virus (strain: 17D); YFV) First, the genetic material coding for structural proteins (capsid protein, prM protein, envelope protein, from 119 to 2452 (2,334 bp)) was prepared from the YFV genome (GenBank: X03700.1) (SEQ. ID NOS. 1 and 2).

[0094] 2) Preparation of genetic material for the varicella-zoster virus (Varicella zoster virus (strain: Oka); VZV)

[0095] The genetic material coding for the gE protein (1,872 bp) was prepared from the VZV genome (SEQ. ID NOs. 3 and 4).

[0096] 3) Preparation of genetic material for the rubella virus (Rubella virus (strain: RA27/3); RuV)

[0097] The genetic material coding for the E2-E1 protein (2,358 bp) was prepared from the RuV genome (SEQ. ID NOs. 5 and 6).

[0098] 4) Preparation of genetic material coding for firefly luciferase (FLuc) (SEQ. ID NO. 15)

(2) Preparation of Vectors (DNA Vaccine Treatments) Expressed in Animal Cells

[0099] The genetic material of YFV was inserted into the gWiz™ vector, which allows protein expression in animal cells. A Kozak sequence (GCCACC) was added at the front of the gene, and a stop codon was added at the end of the gene. The genetic material of VZV was inserted into the gWiz™ vector, with a Kozak sequence added at the front of the gene. The genetic material of RuV was inserted into the gWiz™ vector, with a Kozak sequence (GCCACC) and a start codon (ATG) added at the front of the gene.

[0100] The vector map is shown in FIG. 1. The genes were inserted at the restriction enzyme sites of Sall and NotI.

[0101] The cloned YFV-antigen plasmid (vector backbone: gWiz™), VZV-antigen plasmid (vector backbone: gWiz™), and RuV-antigen plasmid (vector backbone: gWiz™) were transformed into DH5α competent cells. The cells were then cultured in bulk, and the plasmids were recovered using a plasmid prep kit.

(3) Preparation of Auxiliary Genetic Material

[0102] Next, an immunoadjuvant was prepared as an auxiliary genetic material. The gene sequences (SEQ ID NOs. 7 and 8) of the coding region (p35; 127~774 (648 bp), p40; 35~1042 (1,008 bp)) of p35 (GenBank: M86672.1) and p40 (GenBank: M86671.1) of IL-12 were inserted into the pSF-CMV-CMV-Sbfl vector along with the kozak sequence. P35 was inserted into the EcoRI, XhoI sites, and p40 was inserted into the Sall, SpeI sites. The vector map is shown in FIG. 2.

[0103] The cloned IL-12 plasmid (vector backbone: pSF-CMV-CMV-Sbfl) was transformed into DH5α competent cells. The cells were mass-cultured, and the plasmids were recovered using a plasmid prep kit.

(4) Preparation of Vector for In Vitro Transcription (IVT)

[0104] The genetic material of YFV was inserted between the T7 promoter sequence and the bGH poly (A) signal sequence of pcDNA™ 3.1(+) through cloning. The 5' untranslated region sequence and Kozak sequence of the *Homo sapiens* PPARG-related coactivator 1 (PPRC1) gene were added in front of the gene, and the stop codon (TGA), the 3' untranslated region sequence of the *Homo sapiens* alpha-1-globin gene, the poly A sequence, and the Esp3i enzyme cut site sequence were added at the end of the gene. The vector map is shown in FIG. 3.

[0105] The genetic material of VZV was inserted between the T7 promoter sequence and the bGH poly (A) signal sequence of pcDNA3.1™ (+) through cloning. The 5'

untranslated region sequence and Kozak sequence of the *Homo sapiens* PPARG-related coactivator 1 (PPRC1) gene were added in front of the gene, and the 3' untranslated region sequence, poly A sequence, and Esp3i enzyme cut site sequence of the *Homo sapiens* alpha-1-globin gene were added at the end of the gene.

[0106] The genetic material of RuV was inserted between the T7 promoter sequence and the bGH poly (A) signal sequence of pcDNA3.1™ (+) through cloning. The 5' untranslated region sequence, Kozak sequence, and start codon of the *Homo sapiens* PPARG-related coactivator 1 (PPRC1) gene were added in front of the gene, and the 3' untranslated region sequence, poly A sequence, and Esp3i enzyme cut site sequence of the *Homo sapiens* alpha-1-globin gene were added at the end of the gene.

[0107] The genetic material encoding FLuc was inserted between the T7 promoter sequence and the bGH poly (A) signal sequence of pcDNA™ 3.1(+) through cloning. The 5' untranslated region sequence and Kozak sequence of the *Homo sapiens* PPARG-related coactivator 1 (PPRC1) gene were added in front of the gene, and the stop codon, the 3' untranslated region sequence of the *Homo sapiens* alpha-1-globin gene, the poly A sequence, and the Esp3i enzyme cut site sequence were added at the end of the gene.

[0108] The cloned YFV-antigen plasmid (vector backbone: pcDNA™ 3.1(+)), VZV-antigen plasmid (vector backbone: pcDNA™ 3.1(+)), RuV-antigen plasmid (vector backbone: pcDNA™ 3.1(+)), and FLuc plasmid (vector backbone: pcDNA™ 3.1(+)) were transformed into DH5α competent cells. The cells were mass-cultured, and the plasmids were recovered using a plasmid prep kit.

(5) Preparation of Genetic Material (mRNA Vaccine Therapeutics) Expressed in Animal Cells Through IVT

[0109] YFV-antigen plasmid (vector backbone: pcDNA3.1™ (+)), VZV-antigen plasmid (vector backbone: pcDNA™ 3.1(+)), and RuV-antigen plasmid (vector backbone: pcDNA™ 3.1(+)) were used as templates for PCR to prepare linearized DNA (SEQ ID NO: 9, SEQ ID NO: 10, and SEQ ID NO: 11). Primer information is shown in Table 1 below.

TABLE 1

Direction	oligo name	Sequence	
		No.	Remarks
F	F_T7_M4AG	SEQ ID NO. 17	Primers for the PCR (linearized DNA preparation)
R	R_esp3i + 11 bp	SEQ ID NO. 18	(Designed and used a universal primer that can be used regardless of the type of GOI)

[0110] Using the FLuc plasmid (vector backbone: pcDNA™ 3.1(+)) as a template, PCR was performed to induce a frameshift 12 bp downstream of the start codon, and linearized DNA with a stop codon was prepared (SEQ ID NO. 15). The primer information is shown in Table 2 below.

TABLE 2

Direction	oligo name	Sequence No.	Remarks
F	f_fluc	SEQ ID NO. 19	PCR was performed in two steps to induce a stop codon in Flu
R	r_fluc_2	SEQ ID NO. 20	
F	F_t7_fluc_shift	SEQ ID NO. 21	
			1. 1st PCR - template: FLuc plasmid (vector backbone: pcDNA TM 3.1(+); SEQ ID NO. 15) - primers: (F) f_fluc, (R) r_fluc_2
			2. 2nd PCR - template: amplicon of 1st PCR - primers: (F) F_t7_fluc_shift, (R) r_fluc2

[0111] IVT was performed using linearized DNA as a template. Following this, DNase treatment was conducted, and mRNA was precipitated and purified using LiCl. The prepared mRNA was transfected into the Vero E6 cell line to confirm antigen expression. It was confirmed that the FLuc mRNA containing the stop codon did not result in protein expression before it was used.

2. Confirmation of Death of Adipocytes

(1) Experimental Method

1) Diet Induced Obese (DIO) Mouse Production

[0112] C57BL/6J, 3w, female mice were fed a normal diet and a high fat diet with 60% kcal from fat, respectively. Body weight was measured at weekly intervals, and the animals were reared for 10 to 15 weeks to check weight gain before use in the experiment.

[0113] C57BL/6J, 4w, male mice were fed a high fat diet with 60% kcal from fat. Body weight was measured at weekly intervals, and the animals were reared for 10 to 15 weeks to check weight gain before use in the experiment.

2) Induction of Pre-Existing Immunity and Measurement of Antibody Titer

[0114] Pre-existing immune response was induced using attenuated live viruses as vaccine strains. YFV (strain: 17D, GenBank: X03700.1) and VZV (strain: Oka, Sky Zoster strain (SK Bioscience)) were used as the vaccine strains.

[0115] (i) YFV immunization: The YFV vaccine strains (attenuated live virus) was injected into the muscle of the left hind limb of the mouse at a concentration of 2E5 pfu/time 3 times at a 2-week interval. Blood was collected 2 weeks after the last injection, and serum was separated.

[0116] The confirmation of Antibody titer: In the separated serum, antibodies specific to YFV were detected through ELISA analysis. Through ELISA analysis, YFV used for injection was coated on a plate at a concentration of 5E4 pfu/well, and an analysis sample (serum) was diluted and reacted for 2 h. After washing, an anti-mouse IgG-HRP secondary antibody was diluted by 1/4000 and treated, and then reacted for 2 h. After washing, a color forming solution was added and reacted for 10 min to measure an O.D value.

[0117] Results: Normal immune induction (antibody response) occurred by YFV vaccine administration.

[0118] The results are shown in FIG. 4. Compared to the negative control (NC) group administered with medium, an increase in YFV-specific antibodies was confirmed in the YFV-ND and YFV-HFD groups administered with YFV.

[0119] YFV-specific pre-existing immunity induction for treatment administration was confirmed.

[0120] (ii) YFV immunization (2): YFV (attenuated live virus) was administered twice to the left hind limb muscle of the mouse at 2-day intervals 3 weeks before starting treatment administration. (Intramuscular injection, 1st shot: 2E4 pfu/injection, 2nd shot: 4E4 pfu/injection) An additional YFV immunization was administered 2-3 days before treatment administration, i.e. 3 weeks after the 1st shot. (Intramuscular injection, 3rd shot: 1 E5 pfu/injection)

[0121] The confirmation of Antibody titer: Blood samples were collected before YFV administration and 3 weeks after the first shot administration, and serum was separated. For ELISA analysis, YFV used in the injection was coated on the plate at a concentration of 1E4 pfu/well, blocked with 2% skim milk, and then diluted serum was reacted with the antigen for 1 h. After washing, anti-mouse IgG-HRP secondary antibody was diluted 1/1000 and reacted for 1 h. After washing, the color development solution was added and reacted for 10 min, and the OD value was measured.

[0122] Results: Normal immune induction (antibody response) occurred by administration of YFV vaccine strain. The results are shown in FIG. 5. Compared with serum before YFV immunization (pre-vaccination), an increase in YFV-specific antibody was confirmed in serum sampled after two YFV immunizations (post-vaccination). Induction of YFV-specific pre-existing immunity for therapeutic administration was confirmed.

[0123] (iii) VZV immunization: Mice were immunized twice with VZV (attenuated live virus) at 4-week intervals. (Subcutaneous injection, 1st shot: 9E3 pfu/injection, 2nd shot: 9E3 pfu/injection) An additional VZV immunization was performed one week before the administration of treatment. (Subcutaneous injection, 3rd shot: 9E3 pfu/injection)

[0124] The confirmation of Antibody titer: Blood was collected before VZV administration and 3 weeks after the 2nd shot administration, and serum was separated. The separated serum was analyzed by ELISA to detect VZV gE protein-specific antibodies. For ELISA analysis, recombinant VZV gE protein was coated on the plate at a concentration of 100 ng/well, blocked with 2% skim milk, and then the diluted serum was reacted with the antigen for 1 h. After washing, anti-mouse IgG-HRP secondary antibody was diluted 1/1000 and reacted for 1 h. After washing, the color development solution was added and reacted for 10 min to measure the OD value.

[0125] Results: Normal immunity (antibody response) occurred by VZV vaccine administration. The results are shown in FIG. 6 and FIG. 7. Compared to the serum before VZV immunization (pre-vaccination), an increase in VZV gE protein-specific antibodies was confirmed in the serum sampled after two VZV immunizations (post-vaccination).

Induction of VZV-specific pre-existing immunity for therapeutic administration was confirmed.

tissue were separated and their respective weights were measured.

TABLE 3

Group	Classification	Diet 1	Diet 2 (3 days before treatment~autopsy)	Pre-existing immunity induction (3th)	Information of treatment	Number of treatment doses	Number of subjects
Control 1	Normal	ND	ND	Medium	—	—	5
Control 2				YFV vaccine strain	—	—	5
Mock	DIO	HFD		YFV vaccine strain	Empty vector	5 times at 2-day intervals	10
Treatment group 1				YFV vaccine strain	YFV-antigen DNA	5 times at 2-day intervals	10
Treatment group 2				YFV vaccine strain	YFV-antigen DNA, IL-12 DNA	5 times at 2-day intervals	10

[0126] (iv) RuV immunization: The experimental animals were immunized 2-3 times with the MMR vaccine (vaccine containing RuV attenuated live virus) at 4-week intervals. (Subcutaneous injection)

[0127] The confirmation of Antibody titer: Blood samples were collected before MMR vaccination and 3 weeks after the second vaccination, and serum was isolated. The separated serum was detected for RuV E1 protein-specific antibodies through ELISA analysis. For ELISA analysis, recombinant RuV E1 protein was coated on the plate at a concentration of 100 ng/well, blocked with 2% skim milk, and the diluted serum was reacted with the antigen for 1 h. After washing, anti-mouse IgG-HRP secondary antibody was diluted 1/1000 and reacted for 1 h. After washing, the color development solution was added and reacted for 10 min, and the OD value was measured.

[0128] Results: Similar to YFV and VZV administration, induction of RuV-specific pre-existing immunity for therapeutic administration was confirmed.

3) Confirmation of Therapeutic Vaccine Administration and Adipose Tissue Reduction Effect

(i) DNA Vaccine Administration (1) (YFV-Immunized Group)

[0129] Method: Normal mice or mice induced to become obese were divided into groups administered YFV vaccine and groups not administered YFV vaccine, respectively, and used in the experiment. All mice were provided with a normal diet (ND) starting 3 days before the therapeutic vaccine administration. The DNA vaccine (therapeutic agent) was administered to the inguinal (scapula) adipose tissue and inguinal (groin) adipose tissue of the mice. The treatment was administered 5 times at 2 sites on the left or right side of each tissue at 2-3 day intervals. The treatment was administered at a concentration of 100 ug/tissue of YFV-antigen-encoding DNA and 50 ug/tissue of IL-12-encoding DNA, for a total of 150 ug/time, and an empty vector of the same amount as the total amount of DNA vaccine administered was administered to the mock group.

[0130] Body weights were measured before and after the therapeutic vaccine administration. The mice were necropsied 7 days after the last therapeutic vaccine administration. Left/right inguinal adipose tissue and interscapular adipose

[0131] (DIO: Diet-induced obesity, ND: Normal diet, HFD: High-fat diet; YFV-antigen DNA: SEQ ID NO: 1, IL-12 DNA: combination of SEQ ID NOs: 5 and 6)

[0132] Results: The average body weight change rate and adipose tissue reduction effect after treatment administration for each group in Table 3 above were confirmed and shown in FIGS. 8 and 9.

[0133] Referring to FIG. 8, body weight was measured before and after treatment administration, and the body weight change was measured based on the body weight immediately before administration. In the case of the group that was not treated at all (control 1), body weight gradually increased, but the Mock, treatment 1, and treatment 2 groups all showed a body weight reduction effect, and the treatment 1 and treatment 2 groups showed a greater decrease compared to the Mock group. The treatment 1 group also showed an excellent body weight reduction effect, and the treatment 2 group, which was administered with IL-12, showed the greatest body weight reduction. These results show that treatment 1 alone showed an excellent body weight reduction effect.

[0134] FIG. 9 shows the fat weight reduction effect by treatment administration. The treatment was administered to only the left or right side, and on the 7th day after 5 administrations of the treatment, the inguinal AT (adipose tissue) and interscapular AT on the left and right sides of the mice were separated and the AT weight was measured. The results of FIG. 9 are presented by group by calculating the effect of reducing fat weight by comparing the left and right AT weights within the same individual. When comparing the control group (no treatment administered) and the mock group (empty vector administered), it can be seen that the adipose tissue weight was reduced by administering treatment 1 and treatment 2.

[0135] These results show that the treatment of the present invention is not only effective in reducing body weight, but also has the effect of reducing the weight of adipose tissue. Compared to the mock group, the average value of the adipose tissue weight was significantly reduced. It can be seen that the change in adipose tissue weight was much greater in the treatment 2 administration group that was administered together with IL-12.

(ii) Administration of DNA Vaccine (2) (VZV-Immunized Group)

[0136] Method: Among mice induced to be obese by a high-fat diet (HFD), those confirmed to have pre-existing

VZV-specific immunity were selected for the administration of the DNA vaccine (therapeutic agent). Mice were fed HFD until the administration of the therapeutic agent, and from the day of administration, they were alternately fed normal diet (ND) and HFD every three days.

[0137] The treatment was administered to the inguinal adipose tissue of the mice. The agent was injected at four sites on either the left or right side of each tissue, with a total of five administrations at two-day intervals. The treatment was administered at a concentration of 100 μ g/tissue of VZV-antigen-encoding DNA, totaling 100 μ g per administration, and an equal amount of empty vector was administered to the mock group.

[0138] Body weights were measured before and after therapeutic vaccine administration. Mice were necropsied 12-14 days after the last therapeutic vaccine administration. The left and right inguinal adipose tissues were isolated, and their weights were measured.

TABLE 4

Group	Classification	Diet 1	Diet 2 (Day of treatment administration~Autopsy)	Pre-existing immunity induction (3th)	Information of treatment	Number of treatment doses	Number of times treatment is administered	Number of subjects
Control	DIO	HFD	Feed ND and HFD	VZV vaccine strain	—	—	—	9
Mock			alternately every 3 days	VZV vaccine strain	Empty vector	100 μ g/tissue	5 times at 2-day intervals	10
Treatment				VZV vaccine strain	VZV-antigen DNA	100 μ g/tissue	5 times at 2-day intervals	9

(DIO: Diet-induced obesity, ND: Normal diet, HFD: High-fat diet; VZV-antigen DNA: SEQ ID No. 3)

[0139] Results: The average body weight change rate and adipose tissue reduction effect after treatment administration for each group in Table 4 above were confirmed, respectively, and are shown in FIGS. 10 and 11.

[0140] Referring to FIG. 10, body weight was measured before and after treatment administration, and the body weight change rate was calculated based on the body weight immediately before administration. Weight loss was observed in all groups that were alternately fed ND and HFD for 3 days from the day of treatment administration. In particular, a greater weight loss was confirmed in the mock and treatment groups that were administered DNA compared to the control group (no treatment administered). The weight loss rates before administration (day 0) and immediately before autopsy in the mock and treatment groups were similar.

[0141] FIG. 11 shows the fat weight reduction effect by treatment administration. The treatment was administered to only the left or right side, and the weight of the left and right inguinal adipose tissues of the mice was measured on days 12 to 14 after 5 administrations of the treatment. The results in FIG. 11 are presented by group, calculating the rate of adipose tissue reduction by comparing the weight of left and right adipose tissues separated from the same individual. A decrease in adipose tissue was confirmed in the mock group and treatment group that received DNA compared to the control group (no treatment administered). In particular, a greater rate of reduction was confirmed in the treatment group compared to the mock group.

TABLE 5

	Control	Mock	Treatment
1	4.6	-31.9	-44.3
2	12.3	-28.9	-41.3
3	8.6	-20.2	-32.9
4	-12.4	-26.9	-34.3
5	-6.1	-21.4	-23.0
6	0.2	-47.5	-37.2
7	-10.5	-33.2	-44.9
8	-9.7	-30.1	-34.9
9	-12.2	-39.6	-49.8
10	—	-37.1	—
(mean)	-2.80	-31.69	-38.05
(SD)	9.47	8.26	8.02

[0142] As can be seen in Table 5 and FIG. 11, the average value of adipose tissue reduction in the treatment group of the present invention significantly decreased compared to

the Control group. Furthermore, when compared to the Mock group, the degree of adipose tissue reduction was even more impressive. Considering the weight and size of the adipose tissue, the achievement of such a reduction effect is a remarkable outcome.

[0143] (iii) Administration of mRNA Vaccine (3) (YFV-immunized group) Method: Among mice induced with obesity through a high-fat diet (HFD), those confirmed to have YFV-specific pre-existing immunity were selected for mRNA vaccine (treatment) administration. Mice were fed HFD until the administration of the treatment, and from the day of treatment, they were alternately fed normal diet (ND) and HFD every three days. The treatment was administered to the inguinal adipose tissue of the mice. The treatment was given in four spots on either the left or right side of each tissue, at intervals of two days, for a total of five administrations. The treatment was administered at a concentration of 10 μ g/tissue of YFV-antigen-encoding mRNA per session. The mock group received the same total amount of mRNA as the treatment group, consisting of FLuc-encoding mRNA that does not express protein due to the inclusion of a stop codon. Body weight was measured before and after the administration of the therapeutic vaccine. After 12 to 14 days following the last administration of the therapeutic vaccine, the mice were euthanized, and the left and right inguinal adipose tissues were separated and weighed.

TABLE 6

Group	Classification	Diet 1	Diet 2 (Day of treatment administration~Autopsy)	P Pre-existing immunity induction (3th)	Information of treatment	Number of treatment doses	Number of times treatment is administered	Number of subjects
Control	DIO	HFD	Feed ND and HFD alternately every 3 days	VZV vaccine strain	—	—	—	9
Mock				YFV vaccine strain	Fluc mRNA(with stop codon)	10 µg/tissue	5 times at 2-day intervals	9
Treatment				YFV vaccine strain	YFV-antigen mRNA	10 µg/tissue	5 times at 2-day intervals	10

(DIO: Diet-induced obesity, ND: Normal diet, HFD: High-fat diet; treatment: SEQ ID No. 2)

[0144] Results: The average change rate in body weight and the effect of adipose tissue reduction after treatment administration for each group, as shown in Table 6, are illustrated in FIGS. 12 and 13. Referring to FIG. 12, body weight was measured before and after the treatment, and the change rate in body weight was calculated based on the weight just before administration. Weight loss was observed in all groups that alternated between normal diet (ND) and high-fat diet (HFD) every three days from the day of treatment administration. Notably, compared to the control group (no treatment), a greater weight loss was confirmed in the mock and treatment (therapeutic) groups that received the mRNA. In particular, the weight loss rate just before euthanasia after treatment was higher in the treatment group compared to the mock group.

TABLE 8-continued

	Control	Mock	Treatment
9	-12.2	-25.4	-32.6
10	—	—	-37.3
(mean)	-2.80	-13.7	-32.5
(SD)	9.47	13.2	8.0

(iv) Administration of mRNA Vaccine (4) (VZV-Immunized Group)
[0146] Method: Among mice induced to be obese by a high-fat diet (HFD), those confirmed to have pre-existing VZV-specific immunity were selected for the administration of the mRNA vaccine (therapeutic agent). Mice were fed

TABLE 7

(%)	Day 0	Day 2	Day 4	Day 6	Day 8	Day 11	Day 13	Day 15	Day 19	Day 22
control	0.00	-10.17	-7.90	-8.13	-11.91	-5.20	-10.68	-5.75	-7.40	—
mock	0.00	-3.56	0.23	-3.50	-6.71	0.57	-5.15	-1.22	-4.02	-7.81
treatment	0.00	-6.46	-2.23	-6.91	-11.06	-4.15	-9.67	-5.29	-9.17	-11.30

[0145] FIG. 13 illustrates the effect of treatment administration on fat weight reduction. The treatment was administered only to one side (left or right), and after five administrations of the treatment, the left and right inguinal (subcutaneous) adipose tissues of the mice were separated and weighed on days 12 to 14. The results in FIG. 13 show the adipose tissue reduction rate calculated by comparing the weights of the left and right adipose tissues separated from the same individual, presented by group. Adipose tissue reduction was observed in the mock group and treatment (therapeutic) group that received the mRNA compared to the control group (no treatment). Notably, a greater reduction rate was confirmed in the treatment group compared to the mock group. The results are presented in Table 8.

TABLE 8

	Control	Mock	Treatment
1	4.6	6.6	-35.4
2	12.3	-7.0	-28.7
3	8.6	-9.9	-17.6
4	-12.4	3.4	-24.1
5	-6.1	-31.9	-28.3
6	0.2	-14.3	-35.4
7	-10.5	-24.1	-41.9
8	-9.7	-20.8	-43.5

HFD until the administration of the therapeutic agent, and from the day of administration, they were alternately fed normal diet (ND) and HFD every three days.
[0147] The treatment was administered to the inguinal adipose tissue of the mice. The agent was injected at four sites on either the left or right side of each tissue, with a total of five administrations at two-day intervals. The treatment was administered at a concentration of 10 µg/tissue of VZV-antigen-encoding mRNA per administration. The FLuc-encoding mRNA containing a stop codon, which does not express proteins, was administered to the mock group in an amount equal to the total amount of mRNA in the administered therapeutic agent.
[0148] Body weight was measured before and after the administration of the therapeutic vaccine. After 12 to 14 days following the last administration of the therapeutic vaccine, the mice were euthanized. The left and right inguinal adipose tissues were isolated, and their weights were measured.

TABLE 9

Group	Classification	Diet 1	Diet 2 (Day of treatment administration~Autopsy)	P Pre-existing immunity induction (3th)	Information of treatment	Number of treatment doses	Number of times treatment is administered	Number of subjects
Control	DIO	HFD	Feed ND and HFD alternately every 3 days	VZV vaccine strain	—	—	—	9
Mock				VZV vaccine strain	Fluc mRNA(with stop codon)	10 µg/tissue	5 times at 2-day intervals	10
Treatment				VZV vaccine strain	VZV-antigen mRNA	10 µg/tissue	5 times at 2-day intervals	9

[0149] (DIO: Diet-induced obesity, ND: Normal diet, HFD: High-fat diet; VZV-antigen mRNA: SEQ ID No. 4)

[0150] Results: The average weight change rates and the effects of adipose tissue reduction for each group after the administration of the therapeutic agent, as shown in Table 9, were confirmed and are presented in FIGS. 14 and 15.

[0151] Referring to FIG. 14, body weight was measured before and after the administration of the therapeutic agent, and the weight change rate was calculated based on the weight immediately prior to administration. Weight loss was observed in all groups that were alternately fed normal diet (ND) and high-fat diet (HFD) every three days from the day of administration. Notably, greater weight loss was confirmed in the mock and treatment (therapeutic agent) groups that received mRNA compared to the control group (no treatment). The weight loss rates for the mock and treatment groups before administration (day 0) and just prior to euthanasia after administration were similar. The values are presented in Table 10.

HFD until the administration of the therapeutic agent, and from the day of administration, they were alternately fed normal diet (ND) and HFD every three days.

[0154] The treatment was administered to the inguinal adipose tissue of the mice. The agent was injected at four sites on either the left or right side of each tissue, with a total of five administrations at two-day intervals. The treatment was administered at a concentration of 100 µg/tissue of RuV-antigen-encoding DNA, totaling 100 µg per administration, and an equal amount of empty vector was administered to the mock group.

[0155] Body weight was measured before and after the administration of the therapeutic vaccine. After 12 to 14 days following the last administration of the therapeutic vaccine, the mice were euthanized. The left and right inguinal adipose tissues were isolated, and their weights were measured.

[0156] Results: Similar to the YFV and VZV treatment groups, weight loss and adipose tissue reduction effects were confirmed in the treatment administration group.

TABLE 10

(%)	Day 0	Day 2	Day 4	Day 6	Day 8	Day 11	Day 13	Day 15	Day 20
control	0	-10.17	-7.90	-8.13	-11.91	-5.20	-10.68	-5.75	-7.40
mock	0	-5.63	-1.61	-6.18	-10.24	-4.00	-9.74	-6.15	-10.75
treatment	0	-6.44	-1.65	-7.14	-10.77	-3.09	-9.36	-4.89	-10.09

[0152] FIG. 15 shows the effect of weight reduction in adipose tissue due to the administration of the therapeutic agent. The treatment was administered only to either the left or right side, and on days 12 to 14 after five administrations of the therapeutic agent, the left and right inguinal (groin) adipose tissues of the mice were isolated and weighed. The results in FIG. 15 present the adipose tissue reduction rates calculated by comparing the weights of the left and right adipose tissues isolated from the same individuals. A reduction in adipose tissue was confirmed in the mock and treatment (therapeutic agent) groups that received mRNA compared to the control group (no treatment). Notably, a greater reduction rate was observed in the treatment group compared to the mock group.

(v) Administration of DNA Vaccine (5) (MMR-Immunized Group)

[0153] Method: Among mice induced to be obese by a high-fat diet (HFD), those confirmed to have pre-existing RuV-specific immunity were selected for the administration of the mRNA vaccine (therapeutic agent). Mice were fed

[0157] (vi) Administration of mRNA Vaccine (6) (MMR-immunized group)

[0158] Method: Among mice induced to be obese by a high-fat diet (HFD), those confirmed to have pre-existing RuV-specific immunity were selected for the administration of the mRNA vaccine (therapeutic agent). Mice were fed HFD until the administration of the therapeutic agent, and from the day of administration, they were alternately fed normal diet (ND) and HFD every three days.

[0159] The treatment was administered to the inguinal adipose tissue of the mice. The agent was injected at four sites on either the left or right side of each tissue, with a total of five administrations at two-day intervals. The treatment was administered at a concentration of 10 µg/tissue of RuV-antigen-encoding mRNA per administration. The FLuc-encoding mRNA containing a stop codon, which does not express proteins, was administered to the mock group in an amount equal to the total amount of mRNA in the administered therapeutic agent.

[0160] Body weight was measured before and after the administration of the therapeutic vaccine. After 12 to 14 days following the last administration of the therapeutic

vaccine, the mice were euthanized. The left and right inguinal adipose tissues were isolated, and their weights were measured.

[0161] Results: Similar to the YFV and VZV treatment experiments, weight loss and adipose tissue reduction effects were confirmed in the treatment administration group.

TABLE 11

Sequence information		
No.	Explanation	Sequence Number for internal reference and notes
SEQ. ID No. 1	DNA sequence encoding the structural protein of YFV (positions 119-2452 (2,334 bp))	Reference sequence no. 1a DNA sequence encoding the structural protein of YFV (capsid protein, prM protein, envelope protein, 119~2452 (2,334 bp))
SEQ. ID No. 2	mRNA sequence encoding the structural protein of YFV (positions 119~2452 (2,334 bp))	Reference sequence no. 7b YFV-antigen-expressing mRNA prepared by IVT using Reference sequence no. 7a as a template
SEQ. ID No. 3	DNA sequence encoding VZV gE protein (1,872 bp)	Reference sequence number 2a VZV gE protein (1,872 bp) DNA sequence
SEQ. ID No. 4	mRNA sequence encoding VZV gE protein (1,872 bp)	Reference sequence number 8b VZV-antigen-expressing mRNA prepared by IVT using Reference sequence number 8a as a template
SEQ. ID No. 5	DNA sequence encoding the E2-E1 protein of RuV (2,358 bp)	Reference sequence number 3a DNA sequence of E2-E1 protein (2,358 bp) of RuV
SEQ. ID No. 6	mRNA sequence encoding the E2-E1 protein of RuV (2,358 bp)	Reference sequence number 9b RuV-antigen-expressing mRNA prepared by IVT using Reference sequence number 9a as a template
SEQ. ID No. 7	DNA sequence of p35 of IL-12	Reference sequence number 5 IL-12 p35 (GenBank: M86672.1; 127~774 (648 bp))
SEQ. ID No. 8	DNA sequence of p40 of IL-12	Reference sequence number 6 IL-12 p40 (GenBank: M86671.1; 35~1042 (1,008 bp))
SEQ. ID No. 9	Template DNA of sequence number 2	Reference sequence number 7a Linearized DNA prepared by performing PCR using YFV-antigen plasmid (vector backbone: pcDNA3.1 TM (+)) as a template
SEQ. ID No. 10	Template DNA of sequence number 4	Reference sequence number 8a Linearized DNA prepared by performing PCR using VZV-antigen plasmid (vector backbone: pcDNA3.1 TM (+)) as a template
SEQ. ID No. 11	Template DNA of sequence number 6	Reference sequence number 9a Linearized DNA prepared by performing PCR using RuV-antigen plasmid (vector backbone: pcDNA3.1 TM (+)) as a template
SEQ. ID No. 12	Amino acid sequence of the structural protein of YFV	Reference sequence number 1b Structural protein of YFV (capsid protein, prM protein, envelope protein): Amino acid sequence of the expressed protein
SEQ. ID No. 13	Amino acid sequence of the VZV gE protein	Reference sequence number 2b VZV gE protein: Amino acid sequence of the expressed protein
SEQ. ID No. 14	Amino acid sequence of the E2-E1 protein of RuV	Reference sequence number 3b E2-E1 protein of RuV: Amino acid sequence of the expressed protein
SEQ. ID No. 15	Genetic material encoding firefly luciferase	Reference sequence number 4a Firefly luciferase (FLuc) (1,653 bp)
SEQ. ID No. 16	Amino acid sequence of Firefly luciferase	Reference sequence number 4b Firefly luciferase (FLuc): Amino acid sequence of expressed protein
SEQ. ID No. 17	F_T7_M4AG primer	
SEQ. ID No. 18	R_esp3i + 11 bp primer	
SEQ. ID No. 19	f_fluc primer	
SEQ. ID No. 20	r_fluc_2 primer	
SEQ. ID No. 21	F_t7_fluc_shift primer	

INDUSTRIAL APPLICABILITY

[0162] The present invention can prevent and/or treat obesity. The present invention provides a method which can prevent and/or treat obesity by injecting into a localized region.

SEQUENCE LISTING

Sequence total quantity: 21

SEQ ID NO: 1 moltype = DNA length = 2334
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 source 1..2334
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 organism = Yellow fever virus

SEQUENCE: 1

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SEQ ID NO: 2 moltype = RNA length = 2617
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SEQUENCE: 2

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gtgtcaaaag gagccccctg caggattcca gtgatagtag ctgatgatct tacagcgcca 1920
atcaataaag gcatttttgt tacagttaac cccatcgctc caaccaatga tgatgaagt 1980
ctgattgagg tgaaccaccc ttttgagac agctacatta tgcgtgggag aggagattca 2040
cgtctcactt accagtggca caaagaggga agctcaatag gaaagtgtgt cactcagacc 2100
atgaaaggcg tggaaacgct ggcgctcatg ggagacaccg cctgggattt cagctccgct 2160
ggagggttct tcacttcggt tgggaaagga attcatacgg tgtttggctc tgcctttcag 2220
gggtcatttg gcgcttgaa ctggataaca aaggtcatca tgggggcggg acttatatgg 2280
gttgcatca acacaagaaa catgacaatg tccatgagca tgatcttgtt aggagtgatc 2340
atgatgtttt tgtctctagg agtggggcg tgagctggag cctcggtggc catgcttctt 2400
gcccccttgg gctccccca gcccctctc ccttctctgc acccgtaacc cctgtgtctt 2460
tgaataaagt ctgagtgggc ggcaaaaaaa aaaaaaaa aaaaaaaa aaagcatat 2520
gactaaaaaa aaaaaaaa agacgcccgt gatcagc 2580
aaaaaaa aaaaaaaa

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SEQ ID NO: 3      moltype = DNA length = 1872
FEATURE          Location/Qualifiers
source           1..1872
                 mol_type = genomic DNA
                 organism = varicella-zoster virus

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SEQUENCE: 3
atgggaaccg tgaacaagcc tgtggtcgcc gtgctgatgg gctttggcat tatcacaggg 60
accctgcgga tcaccaaccc tgtgctggcc agcgtttcta gatacgacga cttccacatc 120
gacgaggaca agctggacac caacagcgctc tacgagccct attaccattc tgatcacgcc 180
gaattcttct ggggtgaatag agcgaggtct agcagaaagg cctatgacca caactcccca 240
tacattttggc ctctgaacga ctacgacggc ttcttgaaa acgcccacga gcaccacggc 300
gtgtacaacc agggaagagg catcgacagc ggagagagac tgatgcagcc aaccagatg 360
agcgcccagg aggaacctgg agacgacacc ggcattccacg ttatccccc actgaacggc 420
gacgatagac acaagatcgt gaacgtggat cagagacagt acggcgatgt gttcaagggc 480
gacctgaatc ctaaacctca gggccaaaga ctgatcgagg tgcctgggga agaaaatcac 540
cccttcaccc tgcgggcccc tatccagcgg attctacggcg tgcgggtacac agagacatgg 600
tccttctctg catctctgac ctgcacgggg gacgcgcgtc ctgccatcca acacatctgt 660
ctgaagcaca ccacctgctt ccaggatgtg gtctgggacg tggattgcgc cgaaaacaca 720
aaagaagatc agctgcccga gctcagctac agattccaag ggaagaagga ggcgaccag 780
ccctggatcg tgggtgaacac cagcacactg ttcgacgagc tggaaactgga cctctctgag 840
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atgcacctgc agtataagat ccacgaggct ccttttgatc tgcgtctgga atggctgtac 1140
gtgcccctcg accctacctg tcagctatg cggctgtata gcacatgtct gtaccacct 1200
aacgcccctc agtgcttgag ccacatgaac agcggatgta cctttaccag cccgcacctg 1260
gccagagag tggccagcac cgtgtaccag aattgcgagc acgccgataa ctacaccgcc 1320
tactgccttg gaattagcca catggaaccc agctttggcc tgatcctgca cgacggcggc 1380
acaacccctt aattcgtgga cacacctgag agtctgagcg gactgtacgt gttcgtgggt 1440
tacttcaacg gccacgtgga agcgtggccc tacacgtgg tctccacgtt ggaatcttc 1500
gttaatgcca tcgaggaacg gggcttcccc cccacagctg gccagccacc tgctacaaca 1560
aagcccaagg aaatcacacc tgtgaacccc ggcacaagcc cactgctcag atacgcggca 1620
tggaccggcg gcctggccgc cgtggtgctg ctatgcctgg tgatcttctt gatctgcacc 1680
gccagagaaa tgagagtga ggcctaccgg gtggacaaaa gccctacaa ccagagcatg 1740
tactacggcg gcctcctctg tgacgacttc gaggacagcg agtccactga tacagaggaa 1800
gaatttgcca acgcccctcg cggtagccac ggaggaagca gctacaccgt gtacatcgac 1860
aagaccagat ga

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SEQ ID NO: 4      moltype = RNA length = 2152
FEATURE          Location/Qualifiers
source           1..2152
                 mol_type = mRNA
                 organism = varicella-zoster virus

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SEQUENCE: 4
gagagcagcg ctgggtgttc aggggccaag gccaccatgg gaacctgtga caagcctgtg 60

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gtcggcgctgc tgatgggctt tggcattatc acaggcaccg tgcggatcac caaccctgtg 120
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agcgtctacgc agccctatta ccattctgat cagcccgaaat cttcttgggt gaatagaggc 240
gagcttagca gaaaggccta tgaccacaac tcccataca tttggcctcg taacgactac 300
gacggcttcc tggaaaacgc ccacgagcac cagcgctgtg acaaccaggg aagaggcatc 360
gacagcggag agagactgat gcagccaacc cagatgagcg cccaggagga cctggggagac 420
gacaccggca tccacgttat cccacactg aacggcgacg atagacacaa gatcgtgaac 480
gtggatcaga gacagtacgg ccatgtgttc aagggcgacc tgaatcctaa acctcagggc 540
caaaactga tgcagggtgc cgtggaagaa aatcacccct tcacctcgcg gggccctatc 600
cagcggatct acggcgctgc gtacacagag acatgggtct tctgacctg tctgacctgc 660
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cctatgcccc tgatatgac atgtctgtac caccctaacc cccctcagtg cctgagccac 1260
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gaacccagct ttggcctgat cctgcacgac ggccgcacaa cctttaaat cgtggacaca 1440
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taaccgggtg acaaaaagccc ctacaaccag agcatgtact acgcccgcct cctgtttgac 1800
gacttcaggg acagcgagtc cactgataca gaggaagaat ttggcaacgc catcgccggt 1860
agccacggag taagcagcta accgtgtac atcgacaaga ccagatgagc tggagcctcg 1920
gtggccatgc ttcttgcccc ttgggcctcc cccagccccc tcctccctct cctgcaccgc 1980
taccgccgtg gtctttgaat aaagtctgag tggcgggcaa aaaaaaaaaa aaaaaaaaaa 2040
aaaaaaaaag catatgacta aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 2100
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaagagacg ccgctgatca gc 2152

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SEQ ID NO: 5      moltype = DNA length = 2358
FEATURE          Location/Qualifiers
source           1..2358
                 mol_type = genomic DNA
                 organism = Rubella virus

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SEQUENCE: 5
ggcgctcccc aggccttctt ggccggcctg ctgctggctg ctgtggccgt ggccacagct 60
agagccggcc tccaacctag agccgacatg gccgccccac ccacactgcc tcagccccct 120
cgggccacgc gccaacacta cggccaccac caccatcagc tgccattcct gggccacgat 180
ggccaccacg gcgcacactt gaggggtggg cagcacccaca gaaacgccag cgactgtctg 240
cctgggcact ggctgcaggg aggcctggga tgttacaacc tgagcgactg gcaccagggc 300
acccacgtgt gccacaccac gcacatggac ttctggtgcg tggaaacaga cagacccccct 360
cctgccacc cccacactct gacaaccgct gctaactcta ccaccgccgc cacacctgcc 420
accgccccag ctccatgtca gcgcggcctg aacgacagct gtggcggtt tctcagcgcc 480
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cctccttggg agctggtggt gctgaccgcc cggcctgagg acggctggac ctgcagtgga 660
gtgcccggcc acccaggcac cagatgtcca gaactggtgt cccctatggg aagagccacc 720
tgttctctct cttcagccct ttggtggct accgccaacg cctgagcct ggaccagccc 780
ctggccgctt ttgtgctgct ggtcccttgg gtctgatct tcatggtgtg cagaagaaca 840
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cacccccccc gccccctggg cttgaagtgc aagaccgtgc ggcccggtgc cctgctagg 1920
accctggccc ctctcgggaa tgtcgggggt accagtgccg tactcccgt 1980
ctggtggagg gcctcgcccc tggcggaggg aattgtcacc tgaccgtcaa cgagaggac 2040

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ctggggcgccct ttctctcttg caagttctgt accgcccgcg tgetgaatac cccccctcct 2100
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gcccagagct tcaccggagt tgtgtacgga acccacacaa ccgcagttag tgaacaaga 2220
cagacctggg ccgagtgggc ggtgccccat tgggtggcagc tgacctggg cgctatctgc 2280
gccctgctgc tcgctggaact gcttgctgt tgtgccaaat gcctgtacta cctgcgggggt 2340
gccatcgccc caagatga 2358

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SEQ ID NO: 6          moltype = RNA length = 2641
FEATURE              Location/Qualifiers
source               1..2641
                    mol_type = mRNA
                    organism = Rubella virus

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SEQUENCE: 6
gagagcagcg ctgggtgttc aggggccaag gccaccatgg gcgtcccca ggcttctctg 60
gccggcctgc tgetgggtgc tgtggccgtg ggcacagcta gagccggcct ccaacctaga 120
gcccagatgg ccgcccacc cactatgcct cagccccctc gggcccacgg ccaacactac 180
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aggggtggggc agcaccacag aaacgccagc gacgtgctgc ctgggcactg gctgcaggga 300
ggctggggat gttacaacct gagcgactgg caccagggca cccacgtgtg ccacaccaag 360
cacatggact tctgggtgct ggaacacgac agacccctc ctgccacccc cacaccactg 420
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gccggcctga acgacagctg tggcggtctt ctacagcggt gtggccctat gcggctgcgg 540
cacggcgtgc acaccgggtg cggcagactg atttgcggcc tgtccaccac agcccaatac 600
ccccccacca gattcggctg cgcctagaga tggggcctgc ctcttggga gctgggtggtg 660
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ctgcacccgt acccccgtgg tctttgaata aagtctgagt gggcgggcaaa aaaaaaaaaa 2520
aaaaaaaaaa aaaaaaaaaa atatgactaa aaaaaaaaaa aaaaaaaaaa 2580
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaagagacgc cgctgatcag 2640
c 2641

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SEQ ID NO: 7          moltype = DNA length = 648
FEATURE              Location/Qualifiers
source               1..648
                    mol_type = other DNA
                    organism = unidentified

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SEQUENCE: 7
atgtgtcaat cacgtacct cctctttttg gccacccttg ccctcctaaa ccacctcagt 60
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ctgaagacca cagatgacat ggtgaagacg gccagagaaa aactgaaaca ttattcctgc 180
actgtcgaag acatcgatca tgaagacatc acacgggacc aaaccagcac attgaagacc 240
tgtttaccac tggaaactaca caagaacgag agttgcctgg ctactagaga gacttcttcc 300
acaacaagag ggagctgctt gccccacag aagacgtctt tgatgatgac cctgtgcctt 360
ggtagcatct atgaggactt gaagatgtac cagacagagt tccaggccat caacgcagca 420
cttcagaatc acaaccatca gcagatcatt ctagacaagg gcatgctggt ggccatcgat 480
gagctgatgc agtctctgaa tcataatggc gagactctgc gccagaaacc tctgtgggga 540
gaagcagacc cttacagagt gaaaatgaag ctctgcaccc tgcttcacgc cttcagcacc 600
cgctcgtgta ccatcaacag ggtgatgggc tatctgagct ccgcctga 648

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SEQ ID NO: 8      moltype = DNA length = 1008
FEATURE          Location/Qualifiers
source           1..1008
                 mol_type = other DNA
                 organism = unidentified

SEQUENCE: 8
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tggctggtgc aaagaaacat ggacttgaag ttcaacatca agagcagtag cagttcccct 480
gactctcggg cagtgcacat tggaatggcg tctctgtctg cagagaaggt cactctggac 540
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actccccatt cctactcttc cctcaagttc tttgttcgaa tccagcgcaa gaaagaaaag 840
atgaaggaga cagaggaggg gtgttaaccag aaaggtgctg tctctgtaga gaagacatct 900
accgaagtc aatgcaagg cggaatgtc tgcgtgcaag ctcaggatcg ctattacaat 960
tctcatgca gcaagtgggc atgtgttccc tgcagggtcc gatcctag 1008

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SEQ ID NO: 9      moltype = DNA length = 2633
FEATURE          Location/Qualifiers
source           1..2633
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 9
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gtcaaacaaa ataaaaacaaa aaacacacac aattggaaac agacctggac cttcaagagg 180
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ccacctaaag aggttgttga aaatgctgga cccaagacaa ggcttggctg ttctaaggaa 300
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cttggtgcgg aaaaacagat ggttgctcct aaatgtgaca tctgaggacc tgggaaaaac 480
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caggctcagg aggtcaagaa gggccattga cttgcctacg catgaaaacc atggtttgaa 720
gaccgcgcaa gaaaaatgga tgactggaag aatgggtgaa aggcactcc aaaagattga 780
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ggaaacagag agctggatag tggacagaca gtgggcccag gacttgacc tgccatggca 1560
gagtgggaag ggcgggtgtg ggagagagat gcatcatctt gtcgaatttg aacctccgca 1620
tgccgccact atcagagttc tggccctggg aaaccaggaa ggctccttga aaacagctct 1680
tactggcgca atgaggggtt caaaggacac aaatgacaac aacctttaca aactacatgg 1740
tggacatgtt tcttcgagag tgaaattgtc agctttgaca ctcaagggga catcctacaa 1800
aatatgcact gacaaaatgt tttttgtcaa gaacccaact gacactggcc atggcactgt 1860
tgtgatgcag gtgaaagtgt caaaaggagc cccctgcagg attccagtga tagtagctga 1920
tgatcttaca gcggaatca ataaaggcat tttggttaca gttaaccccc tcgctcaac 1980
caatgatgat gaagtgtgta ttgaggtgaa cccacctttt ggagacagct acattatcgt 2040
tggaagagga gattcacgtc tcacttacca gtggcacaaa gagggaagct caataggaaa 2100
gttgttctac cagaccatga aaggcgtgga acgctggcc gtcattggag acaccgctg 2160
ggatttcagc tccgctggag ggttcttcac ttcggttggg aaaggaattc atacggtgtt 2220
tggtctgccc tttcaggggc tatttggcgg cttgaactgg ataacaaagg tcatcatggg 2280
ggcgttactt atatgggttg gcatcaacac aagaacatg acaatgtcca tgagcatgat 2340
cttggttaga gtgatcatga tgtttttgtc tctaggagtt gggcgctgag ctggagcctc 2400
ggtggccatg cttcttgccc cttgggctc cccccagccc ctctctccct tctgcaccc 2460
gtacccccgt ggtctttgaa taaagtctga gtggcgcgca aaaaaaaaaa aaaaaaaaaa 2520
aaaaaaaaaa gcatatgact aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 2580
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaagagac gccgctgatc agc 2633

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-continued

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SEQ ID NO: 10      moltype = DNA length = 2168
FEATURE           Location/Qualifiers
source            1..2168
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 10
taatacgact cactataaga gcagcgctgg gtgttcaggg gccaaaggcca ccatgggaac 60
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gatcaccaac cctgtgctgg ccagcgcttct gagatacgac gacttccaca tcgacgagga 180
caagctggac accaacagcg tctacgagcc ctattaccat tctgacacg ccgaatcttc 240
ttgggtgaat agaggcgagt ctacgagaaa ggcttatgac cacaactccc catacatttg 300
gcctcgtaac gactacgacg gcttctctgga aaacgcccac gagcaccacg gcgtgtacaa 360
ccagggaaga ggcatcgaca gcggagagag actgatgcag ccaaccagga tgagcgcca 420
ggaggacctg ggagacgaca ccggcatcca cgttatcccc acactgaacg gcgacgatag 480
acacaagatc gtgaacgtgg atcagagaca gtacggcgat gtgttcaagg gcgacctgaa 540
tcctaaacct caggggcaaa gactgatcga ggtgtccgtg gaagaaaatc accccttcac 600
cctgcgggcc cctatccagc ggatctacgg cgtgcggtag acagagacat ggtccttcct 660
gcatctcttg acctgcaccg gggacgcccgc tcttgccatc caacacatct gtctgaagca 720
caccacctgc ttccaggatg tggctctgga cgtggattgc gccgaaaaca caaaagaaga 780
tcagctggcc gagatcagct acagattcca agggagaag gaggcggacc agccctggat 840
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tggcgtgctg aaagtgtgta ggaccgagaa acagtacctg ggcgtgtaca tctggaatat 960
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cggccacgtg gaagcctggt cctacaccgt ggtctccacc gtggatcatt tcgttaattg 1560
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cggcctggcc gccgtggtgc tgcctatgct ggtgatcttc ctgatctgca ccgccagag 1740
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cggcctccct gttgacgact ctgagacag cgagtccact gatacagagg aagaatttgg 1860
caacgccatc ggcggtagcc acggaggaaag cagctacacc gtgtacatcg acaagaccag 1920
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cccttctcgt ccccgtagcc cccgtggtct ttgaataaag tctgagtggt cggaacaaaa 2040
aaaaaaaaaa aaaaaaaaaa aaaaaacata tgactaaaaa aaaaaaaaaa aaaaaaaaaa 2100
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa gagacgccgc 2160
tgatcagc                                         2168

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SEQ ID NO: 11      moltype = DNA length = 2657
FEATURE           Location/Qualifiers
source            1..2657
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 11
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cggcctccaa cctagagccg acatggccgc cccaccaca ctgcctcagc cccctcgggc 180
ccacggccaa cactacggcc accaccacca tcagctgcca ttctggggc acgatggcca 240
ccacggcgcg cactgagggg tgggcccagc ccacagaaac gccagcgacg tgctgcttgg 300
gcactggctg cagggagggt ggggatgtta caacctgagc gactggcacc agggcaccca 360
cgtgtgccac accaagcaca tggacttctg gtgcgtggaa cagcagacag cccctcctgc 420
cacccccaca cactgacaaa ccgctgctaa ctctaccacc gccgccacac ctgccaccgc 480
cccagctcca tgtcacgcgc gccctgaacga cagctgtggc ggctttctca gcggctgtgg 540
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caccacagcc caataccccc ccaccagatt cggctgcgcc atgagatggg gctgcctcc 660
ttgggagctg gtggtgtgta ccgcccggcc tgaggacggc tggacctgca gaggagtggc 720
ggccacccca ggcaccagat gtccagaact ggtgtccctc atgggaagag ccacctgttc 780
tcctgcttca gcctgtggc tggctaccgc caacgccctg agcctggacc acgcccgtgc 840
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gacaacagag catccttctc gcaacacccc tcacggccag ttagaggtgc aagtgcctcc 1560

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tgatcctggc gacctgggtg agtacatcat gaaccacaca ggcaaccage agtccagatg 1620
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ctctcctgac tgcagccggc tgggtggggc taccctgtaa agacctagac tgcgcctggt 1740
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cgccacccgt gaaatgcctg agtggattca cgccacacac acaagcgacc cctggcacc 1920
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gccccctctc ccttctctgc acccgtaacc ccgtggtctt tgaataaagt ctgagtgggc 2520
ggcaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa 2580
aaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa 2640
agacgccct gatcagc 2657

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SEQ ID NO: 12      moltype = AA length = 778
FEATURE
source            Location/Qualifiers
                  1..778
                  mol_type = protein
                  organism = Yellow fever virus

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```

SEQUENCE: 12
MSGRKAQGGT LGVNMVRRGV RSLSNKIKQK TKQIGNRPGP SRGVQGFIF FLFNILTGKK 60
ITAHKRLWK MLDPRQGLAV LRKVKRVVAS LMRGLSSRKR RSHDVLTVQF LILGMLMTG 120
GVTLVKKNRW LLLNVTSEDL GKTFVSGTGN CTTNILEAKY WCPDSMEYNC PNLSPREPD 180
DIDWCYGYVE NVRVAYGKCD SAGRSRRSRR AIDLPTHEH GLKTRQEKWM TGRMGERQLQ 240
KIERWFVRNP FFAVTALTIA YLVGSNMTQR VVIALLVLA GPAYSACIG ITDRDFIEGV 300
HGGTVWSATL EQDKCVTVMA PDKPSLDISL ETVAIDRAE VRKVCYNAVL THVKINDKCP 360
STGEAHLAE NEGDNACKRT YSDRGWNGC GLFGKGSIVA CAKFTCAKSM SLFEVDQTKI 420
QVIRAQLHV GAKQENWNTD IKTLKFDALS GSQEVFEFIG GKATLEQVQ TAVDFGNSYI 480
AEMETESWIV DRQWAQDLTL PWQSGSGGVW REMHHLVEFE PPHAATIRVL ALNGEQGSLK 540
TALTGAMRVT KDTNDNNLYK LHGGHVSCRV KLSALT LKGT SYKICTDKMF FVKNPTDTHG 600
GTVVMQVKVS KGAPCRIPVI VADDLTAAIN KGILVTNPI ASTNDDEVL EVNPPFGDSY 660
IIVGRGDSRL TYQWHKEGSS IGKLFTQTMK GVERLAVMGD TAWDFSSAGG FFTSVGKGIH 720
TVFGSAFQGL FGGLNWTIV IMGAVLIWVG INTRNMTSM SMILVGVIM FLSLGVA 778

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SEQ ID NO: 13      moltype = AA length = 623
FEATURE
source            Location/Qualifiers
                  1..623
                  mol_type = protein
                  organism = varicella-zoster virus

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```

SEQUENCE: 13
MGTVNKPVVG VLMGFGIITG TLRITNPVRA SVLRYDDFHI DEDKLDNTNS YEPYHSDHA 60
ESSWVNRGES SRKAYDHNSP YIWPRNDYDG FLENAHEHHG VYNQGRGIDS GERLMQPTQM 120
SAQEDLDGDDT GIHWIPLTNG DDRHKIVNVD QRQYGDVFKG DLNPKPQGQR LIEVSVEENH 180
PFTLRAPIQR IYGVRYTETW SFLPSLTCTG DAAPAIQHIC LKHTTCFQDV VVDVDAENT 240
KEDQLAEISY RFQGGKEADQ PWIVVNTSTL FDELELDPEE IEPGLVKVLR TEKQYLGVIY 300
WNMRGSDGTS TYATFLVTWK GDEKTRNPTP AVTPQPRGAE FHMWNVHSHV FSVGDTFSLA 360
MHLQYKIEHA PFDLLEWLY VPIDPTCQPM RLYSTCLYHP NAPQCLSHMN SGCTFTSPHL 420
AQRVASTVYQ NCEHADNYTA YCLGISHMES SFGLILHDGG TTLKFVDTPS SLSGLVVFV 480
YFNGHVEAVA YTVVSTVDHF VNAIEERGFP PTAGQPATT KPKEITPVNP GTSPLLRVAA 540
WTGGLAAVVL LCLVIFLICT AKRMRVKAYR VDKSPYNQSM YYAGLPVDDF EDSESTDTEE 600
EPGNAIGGSH GGSSYTVYID KTR 623

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SEQ ID NO: 14      moltype = AA length = 786
FEATURE
source            Location/Qualifiers
                  1..786
                  mol_type = protein
                  organism = Rubella virus

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SEQUENCE: 14
MGAPQAFLAG LLLAAVAVGT ARAGLQPRAD MAAPPTLPQP PRAHGQHYGH HHHQLPFLGH 60
DGHGGTTLRV GQHRNADSV LPGHWLQGGW GCYNLSDWHQ GTHVCHTKHM DFVCVEHDRP 120
PPATPTPLTT AANSTTAATP ATAPAPCHAG LNDSCGGFLS GCGPMRLRHG ADTRCGRLIC 180
GLSTTAQYPP TRFGCAMRWG LPPWELVLT ARPEDGWTCR GVPAPGTRC PELVSPMGRA 240
TCSPASALWL ATANALSLDH ALAAFVLLVP WVLIFMVCR TCRRRGAAAA LTAVVLQGYN 300
PPAYGEEAFT YLCTAPGCAT QAPVPVRLAG VRFESKIVDG GCFAPWDLEA TGACICEIPT 360
DVSCGELGAW VPTACARIW NGTQRACTFW AVNAYSSGGY AQLASYFNP GSYKQYHPT 420
ACEVEPAFGH SDAACWGFFT DTVMVFALA SYVQHPHKT VVKFHTETRT VWQLSVAGVS 480
CNVTTEHPFC NTPHGQLEVG VPPDPGDLVE YIMNHTGNQQ SRWGLGSPNC HGPDWASPCV 540
QRHSPDCSRL VGATPERPRL RLVDADPLL RTAPGPGEVW VTPVIGSQAR KCGLHIRAGP 600
YGHATVEMPE WIHAHTSDP WHPPGPLGLK FKTVRPVALP RTLAPPRNVR VTGCYQCGTP 660
ALVEGLAPGG GNCHLTVNGE DLGAPPPGKF VTAALNTNPP PYQVSCGGES DRASARVIDP 720

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SEQ ID NO: 15          moltype = DNA  length = 1653
FEATURE                Location/Qualifiers
source                 1..1653
                        mol_type = genomic DNA
                        organism = unidentified
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SEQ ID NO: 16      moltype = AA  length = 550
FEATURE            Location/Qualifiers
source             1..550
                  mol_type = protein
                  organism = unidentified
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SEQ ID NO: 17      multype = DNA  length = 25
FEATURE            Location/Qualifiers
source             1..25
                  mol_type = other DNA
                  organism = synthetic construct
```

```
SEQ ID NO: 18      multype = DNA  length = 18
FEATURE            Location/Qualifiers
source             1..18
                  mol_type = other DNA
                  organism = synthetic construct
```

```
SEQ ID NO: 19      moltype = DNA   length = 24
FEATURE            Location/Qualifiers
source             1..24
                  mol_type = other DNA
                  organism = synthetic construct
```

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SEQUENCE: 19
atggaagatg ccaaaaacat taag
24

SEQ ID NO: 20      moltype = DNA length = 19
FEATURE           Location/Qualifiers
source            1..19
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 20
ttacacggcg atcttgccg
19

SEQ ID NO: 21      moltype = DNA length = 43
FEATURE           Location/Qualifiers
source            1..43
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 21
taatacgact cactataaga tggaagatgc ctgaaaaaca tta
43

```

1.-14. (canceled)

15. A kit, comprising (a) the pharmaceutical composition comprising at least one virus selected from the group consisting of yellow fever, varicella-zoster virus and rubella viruses; or genetic material encoding a protein derived from thereof; and

(b) an administration guide for administration of the pharmaceutical composition.

16. The kit according to claim **15**, wherein the kit further comprises (c) a composition comprising an antigen for creating an immune environment prior to administration of the (a) composition.

17. The kit according to claim **15**, wherein the (a) pharmaceutical composition further comprises a genetic material encoding a cytokine.

18. A syringe filled with a vaccine composition for size or volume reduction of a target tissue, comprising at least one virus selected from the group consisting of yellow fever, varicella-zoster virus and rubella viruses; or genetic material encoding a protein derived from thereof.

19. The syringe according to claim **18**, wherein the syringe further comprises a genetic material encoding a cytokine.

20. A method for reducing the size or volume of a target tissue of a subject, comprising administering to the target tissue of the subject a genetic material encoding at least one virus selected from the group consisting of yellow fever, varicella-zoster virus and rubella viruses.

21. The method according to claim **20**, wherein the method induces reduction or death of the number of adipose cells or the size of adipose tissue of the subject.

22. The method according to claim **20**, wherein the method comprises

- i) administering to a subject for reducing or shrinking the size of adipose tissue, or killing adipose tissue a vaccine for preventing at least one virus selected from the group consisting of yellow fever, varicella-zoster virus and rubella viruses,
- ii) confirming the production of antibodies by the vaccine, and

iii) administering a pharmaceutical composition comprising at least one virus selected from the group consisting of yellow fever, varicella-zoster virus and rubella viruses; a protein derived from the virus; or the genetic material encoding the virus or the protein.

23. The method according to claim **21**, wherein the method further comprises iv) administering at least one cytokine selected from the group consisting of IL-12, IL-2, IL-4, IL-5, IFN- γ , IL-10, IL-1, IL-6, INF-alpha, INF-beta, TNF-alpha, and TNF-beta simultaneously or sequentially with the pharmaceutical composition.

24. (canceled)

25. (canceled)

26. The method according to claim **20**, wherein the genetic material is polynucleotide encoding at least one protein selected from the group consisting of a structural protein of yellow fever virus, glycoprotein E of varicella-zoster virus, and E2 and E1 proteins of rubella virus.

27. The method according to claim **26**, wherein the polynucleotide is DNA or RNA.

28. The method according to claim **20**, wherein the genetic material comprises a nucleic acid sequence consisting of a polynucleotide selected from the group consisting of SEQ ID NOs: 1 to 6.

29. The method according to claim **20**, wherein the genetic material causes an intracellular immune response.

30. The method according to claim **20**, wherein the genetic material locally acts, and

the genetic material is applied after at least one antigen for creating an immune environment is first applied to a subject.

31. The method according to claim **20**, wherein the genetic material is administered to target tissue orally, by injection through transdermal, intramuscular, peritoneal, intravenous, subcutaneous or nasal route, or by electroporation, gene gun, liposome, dendrimers, nanoparticles, or transfer vectors.

32. The method according to claim **20**, wherein the dose of the genetic material is 0.1-1,000 ug/site per one time inoculation.

* * * * *